

Responsible use of AI in Evidence Synthesis (RAISE 2026) 3: selecting and using AI evidence synthesis tools

Contributing authors and other author details: as per recommendations paper ('RAISE 1')

How to cite

Thomas J, Hair K, Noel-Storr, A. et al. Responsible use of AI in evidence Synthesis (RAISE): selecting and using AI evidence synthesis tools (version 3.1; updated 13 March 2026). In: Open Science Framework [<https://osf.io/>], Washington DC: Center for Open Science. DOI 10.17605/OSF.IO/FWAUD (accessed XXX via <https://osf.io/cqa82>).

Abstract

The evidence synthesis community is being presented with a rapidly growing range of AI tools, each promising to streamline the review process. However, deciding when and how to use AI, and how to document its use transparently, is not straightforward. Technical, ethical and organisational challenges mean that evidence synthesists need to understand key risks and requirements before making informed decisions about adopting AI. The existence of AI alone does not justify its use. Misuse can not only hamper the evidence synthesis process but also undermine the credibility of the work and introduce or exacerbate harms. The RAISE project aims to address these challenges and is writing guidance, currently split into three papers.

The first paper, 'RAISE 1', provides tailored recommendations for eight of the distinct roles in the evidence synthesis ecosystem: evidence synthesists, methodologists, AI tool development teams, organisations that produce evidence synthesis, publishers, funders, users, and trainers of evidence synthesis methods.

RAISE 2 provides guidance on building and evaluating AI tools for evidence synthesis, focusing on assessing whether a tool performs as intended, including how to build and validate tools, evaluate their performance, and report evaluations.

This paper, 'RAISE 3', aims to offer guidance on selecting and using AI evidence synthesis tools. While the field continues to evolve rapidly, we outline the current state of AI in evidence synthesis and provide guidance on assessing tools, including internal and external validity and key ethical, legal, and regulatory considerations.

Highlights

What is already known

- There is considerable excitement about the potential for AI to make the evidence synthesis process more efficient, but also noticeable alarm over the potential for harm
- Guidance is needed to help evidence synthesis authors decide when and how a given tool might be utilised, and when it should not be

What is new

- This paper provides guidance on selecting, using, and transparently reporting AI tools in evidence synthesis evaluations
- The three RAISE papers provide recommendations tailored for tool developers, evidence synthesists, methodologists, funders, organizations that produce evidence synthesis papers, and other members of the research community
- RAISE is designed to help the community work collectively to build AI tools, critically evaluate tools, and ultimately develop an evidence base supporting reliable use of AI in evidence synthesis

Potential impact for RSM readers

- Use of RAISE to guide AI tool development and evaluation should result in more transparent and responsible use of AI in evidence synthesis

Introduction

This is the third paper in the ‘RAISE Guidance’, which aims to support the responsible use of AI in evidence synthesis. The other papers include *Recommendations for Practice*, which provide detailed recommendations for each role on the expertise needed to perform the role effectively, and guidance on *Building and Evaluating AI tools for evidence synthesis*.

This paper offers guidance on *selecting and using AI evidence synthesis tools*. It includes:

- A summary of the current state of AI tools
- Guidance on selecting and using an AI tool
- A framework to assist evidence synthesists or organisations in assessing AI tools for use
- Ethical, legal and regulatory considerations of using AI tools

1: The current state of AI tools

Who is this section for?

- Evidence synthesists
- Evidence synthesis methodologists
- Trainers of evidence synthesis methods

Classes of AI tools and technology

As described in *Building and Evaluating AI tools for evidence synthesis* ('RAISE 2'), a diverse range of AI technologies exists for various evidence synthesis tasks. These are used in both stand-alone tools for individual tasks and in systematic review platforms that provide "end-to-end" AI-driven assistance for evidence synthesis. Several recent reviews provide comprehensive overviews of available tools, focusing on specific tasks and the capabilities of generative LLMs.¹⁻⁵ Understanding the strengths and limitations of the technologies underpinning a given tool aids in assessing how it might (or might not) be used responsibly.

Table 2 provides a current (February 2026) state of AI tools categorised by task and technology class. As stated in RAISE 2 Section 1, the classes are: **rule-based algorithms** (which use fixed rules to make decisions), **unsupervised classifiers** (no labelled training data), **supervised classifiers** (built on labelled training data), **generative LLMs** (that generate text or responses), and **agentic AI** (end-to-end processes which perform multiple review steps).

Writing a protocol

There has been limited progress developing AI tools to support protocol development. Most efforts have focused on developing web interfaces (either as standalone tools or integrated into evidence synthesis platforms) to support protocol completion in accordance with a set of pre-specified rules, often with specific guidance. However, a recent study evaluated the performance of various generative LLMs in assessing adherence to PRISMA guidelines with promising results⁶. While still exploratory, this could pave the way for AI-supported protocol checks to supplement human scrutiny. Researchers have also explored the potential use of generative LLMs to generate novel evidence synthesis questions that could, in the future, lead to the development of tools which identify new research avenues and reduce overlapping synthesis efforts.⁷

The Search

Building and executing an appropriate search strategy to retrieve relevant literature is time-consuming. It has benefited from a wealth of automation technologies, ranging from rule-based algorithms that translate search queries to generative AI tools that refine them.

One approach to support search strategy development is to perform frequency analyses on keywords or controlled vocabularies retrieved from preliminary searches. This can be implemented as part of the search strategy development process to support the expansion of search strategies and ensure comprehensive term coverage. While such tools can improve efficiency, they do not replace the expertise of information specialists and achieve the highest sensitivity when used as a complementary approach alongside expert input.⁸

The use of generative LLMs to generate search terms is a more recent development (but should still be considered as an early starting point, for scoping purposes, or as a supplementary approach.⁹ At present, these tools are only suitable for exploratory use and do not have the capabilities to replace traditional search development methods.

The process of retrieving literature can also be accelerated. Some evidence synthesis platforms, reference managers, and standalone systems utilise application programming interfaces (APIs) to automate the retrieval of studies from common bibliographic databases. Developments in “Agentic AI” – where generative LLM “agents” make plans, execute actions, and make decisions (often with human oversight) - have introduced a host of interfaces which allow users to input natural language queries (e.g. “What is the effect of mindfulness on smoking cessation?”) and retrieve relevant publications. This is made possible through a paradigm shift in retrieval methodology:

- Traditional literature searches rely on manually **curated keywords and phrases**, often combined with Boolean operators. This may involve searching controlled vocabularies (e.g. MeSH terms) or **free-text searching** of the titles and abstracts of indexed publications. The benefits of this approach are that, at least in theory, the queries are reproducible and transparent i.e. it is clear why a certain study was identified or why it was missed.
- **Vector searching** involves retrieving semantically similar items by comparing their numerical vector representations, rather than relying on exact term matches. In this case, language models convert studies from the literature into numerical representations which capture the meaning and context of the text, called vectors.

Agentic AI search platforms (e.g., Elicit, Undermind, Asta Paper Finder) typically use vector representations of the user query to construct a search strategy, execute it in either internal or external bibliographic databases, retrieve a list of potentially relevant records, and rank them by relevance before presenting them to the user. This can be beneficial at the scoping stage of a review, to become more familiar with the terminologies and types of literature available. However, it is not always clear where studies were retrieved from or whether the search was truly comprehensive. Often, the search results are substantially lower than what would be expected from standard bibliographic database searches, as they have been pre-filtered into a shortlist before being returned to the user. For example, searching via Elicit has been found to retrieve fewer than 40% of relevant articles (compared with a sensitivity of 94.5% for traditional systematic reviews).¹⁰ At present, such platforms are most appropriate when used as a supplementary approach to identify studies which traditional searches may have missed.

The inability to comprehensively access all publications and other record types is a key barrier limiting the potential of fully automated retrieval. Many journals remain behind paywalls or are only partially indexed in bibliographic databases. This creates inequities, where researchers at institutions with fewer subscriptions are unable to access portions of the literature. For AI systems, this means that parts of the evidence base cannot be used, either for model training or for efficient retrieval via APIs. This, in turn, may result in incomplete or biased outputs.

Deduplication

There is now a host of mature, validated methods to fully or semi-automate the removal of duplicate records retrieved from different sources.^{11,12} Most algorithms have been developed using rule-based

approaches, with set criteria for how similar the metadata or bibliographic fields (e.g. DOI, title, authors) must be between two candidate matches.

Deduplication tools are packaged in three main formats: standalone tools, integrated into reference managers, or integrated into evidence synthesis platforms. Given the ubiquity of available approaches, it would be unusual to conduct evidence synthesis projects without some degree of automated deduplication.

Screening search results

Automating the task of study selection has received the most attention^{13,14}. Popular systematic review platforms (e.g. DistillerSR, Covidence, EPPI Reviewer, Abstrackr, Rayyan) have integrated AI-assisted screening tools.¹⁵ However, some of these solutions do not have publicly available source code, provide insufficient information about how the classifiers are trained, and lack published performance evaluations.

Most screening tools utilise supervised machine learning approaches, requiring users to manually screen a proportion of articles to provide “training” data for the machine to learn from, as well as unseen “test” data to evaluate machine performance. Another implementation is “prioritised screening”, in which references are sorted by similarity to manually included or excluded references. By design, this enables reviewers to encounter relevant studies earlier in the process, with most excluded studies falling to the bottom of the pile. In theory, reviewers will eventually capture all (or most of) the relevant evidence and can stop screening. Currently, there is some uncertainty regarding when it is ‘safe’ to stop screening, though recent work is making considerable progress.^{16,17} Some teams manually screen everything, adapting their review processes to begin working on eligible records that are identified quickly while screening continues. Others cease screening before all records have been examined, relying on the prioritisation algorithm to have identified the relevant records prior to stopping. While prioritised screening approaches have been used for many years, there is inherent risk in applying stopping criteria without communicating the uncertainty and implications of this approach.¹⁶ Recent empirical evidence evaluating stopping criteria across a range of datasets suggests they are largely overconfident and fail to prevent missing relevant records.¹⁷ Although safer approaches exist, they do not necessarily save the reviewer much time if they aim to prioritise recall and want to be highly confident that the prediction from the stopping algorithm is correct.¹⁸ Callaghan’s algorithm prioritises both accuracy and confidence in the algorithm. Reviewers might choose to sacrifice confidence in the prediction for a reduced workload.^{17,18} This is currently an active topic of discussion in the field.

In recent years, there has been an increasing interest in using generative LLMs for such tasks. With well-constructed prompts¹⁹ and a few example classifications,²⁰ LLMs show promise for citation screening without needing significant amounts of training data.²¹ Efforts are underway to comprehensively evaluate their capabilities versus traditional human screening and determine whether semi-automated (the LLM acts as a single reviewer or reconciler) or fully automated screening performs at a sufficient standard.²²

Data extraction

There is growing interest in the potential of AI tools to address more complex data extraction tasks, in part due to the promise of huge efficiency gains, and in part because the technology is now at a

stage where it might be able to perform these tasks to an acceptable level. In healthcare settings, this often involves highlighting or extracting PICO (population, intervention, comparator, outcome) information from abstracts or full texts. Similarly, in public health, the PECO (population, exposure, comparator, outcome) information is targeted. Approaches include recognising entities using natural language processing and deep-learning techniques, which require domain-specific training data. More high-quality evaluations are needed, but there are signs that carefully designed 'LLM assisted' workflows might be approaching human levels of accuracy, though may not always save as much money as might be expected.²³

Rule-based algorithms may use regular expressions to identify sequences of characters in text corresponding to specific data elements. In work to create living evidence summaries of biomedical research, dictionaries of regular expressions relating to disease models, age, sex, intervention, and outcome measures have been applied to support accelerated synthesis.²⁴ This approach, while helpful for signposting relevant studies, is likely insufficient to extract data with a high degree of accuracy due to extensive variations in terminology, complex experimental designs, and mentions of previous research alongside novel findings. There has been increasing interest in using deep-learning approaches, such as transformer architectures pre-trained on relevant scientific texts.^{25,26} Often, a combination of rule-based and deep-learning approaches is used.^{27,28} Typically, tool evaluations are performed on domain-specific datasets under strict conditions. Any application to a new evidence synthesis project requires context-specific evaluation (e.g. a 'Study Within A Review' or 'SWAR'²⁹) to estimate accuracy.

A challenge lies in the relative scarcity of tools available to reviewers to use (with a user interface), particularly if they lack the technical expertise to implement the source code. Generative LLM approaches allow researchers to experiment with automated data extraction without necessarily requiring a data science skillset and without the need for high volumes of training data. However, the effort required to evaluate LLM outputs versus human extractions for validation purposes is not trivial and requires careful consideration of the types of errors identified.³⁰

Critical appraisal

Automating critical appraisal has often proved more challenging than other stages of the evidence synthesis workflow. The manual application of risk of bias using tools relies heavily on personal judgement, which is often reflected in low inter-rater reliability. This may complicate the creation of gold-standard datasets for building and evaluating AI systems. There is also a multitude of different risk of bias tools for different study designs and domains.

Rule-based regular-expression approaches have been applied to estimate reporting of measures reducing the risk of bias with some success.^{31,32} However, this is less useful at the level of an individual research study or experiment i.e. the accuracy required for critical appraisal as part of an evidence synthesis. Deep learning methods, in which assessments by specific tools are treated as classification tasks, have shown some promise for automating assessment with critical appraisal tools. In an evaluation of RobotReviewer, a system using supervised deep learning methods, humans accepted 83% of the AI systems assessments.³³ A recent comparison of 11 automated tools across 9 different measures of scientific rigour demonstrated that ensemble approaches (where the output scores from multiple tools are considered together) can, in some cases, outperform any individual tool, particularly when there are multiple modalities to consider (e.g. text, images, figures).³⁴

Prompting generative LLMs with detailed instructions for risk of bias assessment has yielded encouraging results, yet careful consideration of how the LLM should handle incomplete information and exercise caution in their judgements based on highly influential words (e.g. “random”). If evaluated in context, it may be acceptable to use LLMs to support bias assessment alongside human checking.^{35–37}

Synthesis

Compared with other evidence synthesis tasks, the automation of data synthesis and analysis appears under-utilised.^{13,38} Several “point and click” tools support the automation of meta-analysis when effect size estimates and summary statistics are available.³⁹ Such platforms are typically rule-based, with underlying R or Python code designed to perform calculations and/or visualise results based on user-defined parameters. In assessing the trustworthiness of outputs from these tools, it is important to examine the quality of the imported data and whether users have the analytical knowledge to determine which analyses are appropriate. It is essential to ensure that analyses can be documented transparently, allowing others to determine whether they make sense in the context of the synthesis question. Other rule-based tools permit users to perform “live” meta-analysis of collected datasets or update an existing analysis based on the addition of new evidence.⁴⁰ The decision of whether such updates are appropriate or not remains with humans. An emerging development is the application of generative LLM chatbots to help users understand meta-analysis concepts and interpret results (e.g. MetaMar^a).

Generative LLMs have also been employed to write and execute analysis code based on extracted data; however, to achieve adequate performance, example code was provided in the prompt, and sufficient technical expertise is required to trust the outputs.⁴¹

Generative LLMs have also been integrated into qualitative assessment tools (e.g., NVivo), where they summarise text or suggest codes to support theme identification and sentiment analysis. To uphold quality standards, all suggestions provided should be checked before being incorporated into the analysis.

Finally, it is critical to bear in mind that synthesis is not the same as text summarisation. Some tools ‘summarise’ results across studies in a ‘synthesis-like’ manner, but they are not doing an evidence synthesis in a methodologically robust way. When an evidence synthesist combines the results from studies in e.g. a meta-analysis, they: consider whether the studies *can* be put together in a valid way; transform statistical results into a standardised form, calculate the magnitude and precision of their combined effects in a way that takes account of the relative size of the studies; and consider heterogeneity, run sensitivity analyses, and generally test the robustness and validity of their analysis. LLMs do none of these things when summarising text, and so therefore we cannot consider current LLM summaries to be equivalent to synthesis.

Assessing certainty in the synthesised evidence

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) is a widely used framework for rating of evidence to inform future health recommendations. The GRADE

^a <https://www.meta-mar.com>

working group has recently unveiled a living online book,^b which will soon replace the GRADE handbook. While the interface aims to help users navigate through the latest methodological guidance and best practices, there is also an ambition to implement a generative LLM companion with additional training on the GRADEbook content to answer specific questions in an interactive fashion. The group has also published some general principles of using AI in health guidelines.⁴²

Automated approaches have also been developed, such as EvidenceGRADER - a deep learning model that assesses study quality in line with the GRADE criteria⁴³ - and URSE – which utilises a number of supervised machine learning tools in a pipeline.⁴⁴ However, in the case of EvidenceGRADER, significant disparities in predictive performance were identified across research domains, and URSE is described as ‘proof-of-concept’. At present, such tools cannot be applied independently without stringent manual checking, which limits efficiency gains.

Text Generation

There is great excitement about the potential for generative AI systems to summarise text accurately. As mentioned above though, merely summarising text is not the same as doing a synthesis, and we should not confuse the two. Having said that, there are evidence synthesis tasks that summarisation might be useful for, though prior research has shown that summarising information reliably across multiple documents is challenging.⁴⁵ Important information can be overlooked while other issues may be overemphasised. Summaries of findings from multiple studies may be particularly challenging, though limited summarisation of specific elements within individual reports (for example, descriptions of interventions) may already be accurate enough for some purposes but outputs still require careful checking by humans.

Generative LLMs have also demonstrated effectiveness in generating plain-language summaries, that convey the key findings of an evidence synthesis in simplified terms.^{46,47} With human oversight, this may encourage reviewers to share findings in alternative, more accessible formats with less upfront effort. The field is moving very quickly though, so we urgently need more evaluations of the newest models as well as methods that make evaluations scalable and sustainable.

It has long been known that many evidence syntheses do not include research reported in languages other than English, resulting in potential bias.⁴⁸ Using automated machine translation has been proposed as one solution to this problem. Relatively few studies evaluate the accuracy of machine translation of research for use in evidence syntheses, though those that do exist have found some promising results.^{49,50} There is some evidence that study design and other easier concepts can be translated accurately, though outcomes and statistical results less so.

Agentic tools which combine multiple stages into a single process

Commonly used evidence synthesis platforms have integrated AI tools to support with different evidence synthesis tasks. For example, EPPI Reviewer offers users a rule-based automated deduplication algorithm for search results, supervised classifiers to support study screening, and generative LLMs to support data extraction. For evaluation purposes, these should each be treated as separate AI tools, with a nuanced understanding of their performance on specific tasks.

^b <https://book.grade.pro.org>

Other AI platforms aim to provide an “end-to-end” interface for conducting literature searches, extracting data, and synthesising findings. Under the hood, inter-connected AI ‘agents’ are performing different tasks, with varied degrees of accuracy. The autonomous nature of these platforms means that the generative LLM agents are making decisions about how to proceed, often with limited transparency. While humans can typically provide feedback or augment decision-making, these “black box” architectures do not present enough information needed to produce reproducible synthesis results. Errors in any single task – for example, a search strategy that misses relevant studies, overly-restrictive screening criteria, or automated data extraction with limited accuracy – can compound, impacting the results presented. It is difficult to evaluate black-box generated reports, without knowledge of the subtasks having been performed. The reliability of LLM ‘syntheses’ of findings across studies is questionable (particularly given many systems are not underpinned by systematic or transparent methods). Therefore, use of LLMs to synthesise findings across studies should be avoided; but remains an important topic for further research, particularly in for Agent tools.

Current recommendations for using AI tools

Table 1 summarises a brief taxonomy of how a given AI technology may be used, and should be used with Table 2 which contains a list of evidence synthesis tasks along with the recommendation for how each class of technology may (or may not) be used. Many technologies are already widely used in evidence synthesis and are generally accepted for use in reviews. A less well-established and generalisable evidence base exists for many other tools, mainly those based on newer generative LLM technology. Here, it may be possible to validate them within the review that they will be used in (see RAISE 2, Section 2), but their use is unlikely to be justifiable without validation. Some tools are only considered safe to use in exploratory phases of the review, or in ways that supplement, rather than replace, conventional review practices. The only technology for which it is hard to see a valid use currently is when LLMs are used to synthesise findings *across* studies; such outputs should be considered unreliable and should not be used in any formal review context.

Table 1: Tool use recommendations

Recommendation	
Acceptable for use	AI outputs may be used directly within the review workflow, if any limitations or potential biases are acknowledged and accounted for.
Human verification required	AI outputs may be used to support review tasks but must be carefully checked by humans before use. The degree of checking required may vary, but typically this will require a human to read and possibly make amendments to the entirety of the output.
Requires validation within the review	AI outputs may be used if their performance is explicitly evaluated within the context of the review itself and deemed adequate (e.g. comparable to human performance).
Exploratory and supplementary use	AI outputs may be used for developing ideas or as a starting point to support understanding. All outputs

	should be extensively refined by human reviewers prior to use for a review task. Alternatively, outputs may be appropriate for use as an additional, supplementary approach, but without replacing established processes.
Not acceptable for use	The current state of technology means that these AI outputs have such serious limitations, that they should not be relied upon.

Table 2: current (February 2026) state of AI tools

Task	Tool Class	Detail and considerations	Example tools	Recommendation
Writing a protocol				
Question formulation	Generative LLMs	Asking LLMs to provide novel questions for synthesis may support early question development. However, suggestions may be incomplete, irrelevant, subject to bias (based on its sources), or overlap with past reviews.	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Human verification required
Drafting	Generative LLMs	Pre-trained LLMs can provide an outline using well-established protocol formats. Users may also provide a format / direct the LLM to resources to support this.	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Human verification required
The Search				
Exploring the literature	Unsupervised	Topic modelling tools aid in identifying clusters of evidence quickly to get a sense of key themes/areas of interest.	Carrot2 (https://search.carrot2.org)	Acceptable for use
	Agentic AI	AI agents develop, refine, and perform searches based on natural language queries. Highly dependent on data sources the tool has access to and requires human input at each stage to guide agent. May be helpful to gain a sense of the literature at an early stage but should not be used as part of any formal evidence retrieval.	Undermind (https://www.undermind.ai/), Elicit (https://elicit.com/), Asta Find Papers (https://asta.allen.ai/)	Acceptable for use
Search strategy development	Rule-based	Tools analyse frequency of keywords and/or controlled vocabularies in search results. Specialised tools are required to cover indexing from different bibliographic databases. May provide additional keywords to inform search strategy but should be used in combination with other search development methods.	Yale MeSH Analyzer (https://mesh.med.yale.edu/), TERA WordFreq (https://tera-tools.com/word-freq); PubReMiner (https://hgserver2.amc.nl/cgi-bin/miner/miner2.cgi),	Acceptable for use

			SearchBuildR (https://pmc.ncbi.nlm.nih.gov/articles/PMC11795901/)	
	Generative LLMs	Highly dependent on prompt quality and database selection. Current evidence suggests that generated strategies are not adequate and require information specialist input. May be more appropriate for early drafts/scoping.	ChatGPT, CoPilot, Claude, Gemini	Exploratory and supplementary use
	Agentic AI	Iterative "agentic" workflow to develop a search strategy with internal and user feedback.	NestedKnowledge Smart Search Tool	Exploratory and supplementary use
Citation chasing	Rule-based	Useful for identifying studies missed by keyword searches, especially in poorly indexed domains. Highly dependent on citation database coverage. May propagate citation bias (i.e., older or more cited work retrieved more often).	CitationChaser (https://estech.shinyapps.io/citationchaser/) ; Connected Papers (https://www.connectedpapers.com/); ResearchRabbit (https://www.researchrabbit.ai/)	Acceptable for use
Citation retrieval	Rule-based	Can be run programmatically or accessible via some evidence synthesis platforms. Not all databases have API access.	TERA, EPPI Reviewer, Nested Knowledge	Acceptable for use
	Agentic AI	AI agents develop, refine, and perform searches based on natural language queries. Highly dependent on data sources the tool has access to and requires human input/verification at each stage to guide agent. Note that these systems are not yet sensitive enough to replace traditional search strategies and retrieval sources but could aid in identifying additional studies.	Undermind (https://www.undermind.ai/), Elicit (https://elicit.com/), Asta Find Papers (https://asta.allen.ai/)	Exploratory and supplementary use

Query translation	Rule-based	Literature databases may change over time, which can break tool functionality. Many tools are not actively maintained or validated. Specialist database coverage may be limited.	TERA Polyglot (https://tera-tools.com/polyglot)	Human verification required
	Generative LLMs	Generative models may be trained in the syntax of prominent bibliographic databases and can be useful for translating	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Human verification required
Query validation	Rule-based	Users input search results and can check retrieval coverage for a reference set of articles. Tools may include matching/deduplication algorithms to perform this task or match on unique identifiers.	TERA Search Refiner, CiteSource	Acceptable for use
Duplicate removal	Rule-based	Logic may be tailored to certain domains and not generalise, so it's important to check validity for a given use case. Most validated tools have low false positive rates. Ideally, tools should offer the capability to download duplicate pairs for transparency.	TERA Deduplicator, ASySD, Deduklik	Acceptable for use
Screening for relevance				
Study selection	Rule-based	Manually crafted, domain specific regular expression dictionaries can be helpful to identify studies with characteristics (particularly when applied to full texts). Dictionary-based tools cannot adapt to new terminology.	Domain specific PICO dictionaries (e.g. in Systematic Online Living Evidence Summaries)	Need to validate within review
	Supervised prioritised screening algorithms	To preserve sensitivity, challenges around decision of when to stop screening. Evidence suggests that time savings may be minimal for maximal sensitivity. Most tools trained only on title/abstract text.	EPPI Reviewer Prioritised Screening; Abstrackr	Need to validate within review
	Supervised classifiers	Performance depends heavily on relevance/quality of training and validation data. Amount/diversity of data needed to train a new classifier varies by concept complexity. Most tools trained only on title/abstract text.	EPPI Reviewer Supervised Classifiers; Abstrackr	Need to validate within review

	Generative LLMs	Capabilities still emerging. May be insufficient for complex inclusion/exclusion criteria without additional training or structured examples.	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Need to validate within review
Quality assessment				
Reporting quality	Rule-based	Useful where terminology is consistent and structured. Best suited as decision prompts rather than definitive judgments. May be helpful for rapid assessments at scale or study selection (see above)	Regex-based assessment of key reporting quality measures for preclinical studies	Need to validate within review
	Supervised classifiers	Performance will be highly dependent on training corpus and domain.	STEED (https://github.com/Ineichen-Group/Auto-STEED); Pre-ROB (https://github.com/camaradesuk/pre-rob)	Need to validate within review
Risk of bias assessment	Supervised classifiers	Performance will be highly dependent on training corpus and domain.	RobotReviewer, SciScore	Need to validate within review
	Generative LLMs	Risk of hallucination is particularly relevant because reporting is often incomplete or ambiguous. Careful prompting and verification required.	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Need to validate within review
Data extraction				
Named entity extraction	Supervised classifiers	Frequently used for PICO/PECO entity extraction. Performance dependent on domain and training corpora.	RobotReviewer (https://pubmed.ncbi.nlm.nih.gov/26104742/), Dextr (https://pubmed.ncbi.nlm.nih.gov/34920276/)	Need to validate within review

	Hybrid	Combination of approaches to achieve optimal performance. For example, a tool may apply rule-based methods to detect candidate entities, then apply machine learning to filter results.	Examples combining rule-based and generative LLM methods for PICO extraction from biomedical literature (https://aclanthology.org/2024.bionlp-1.39/);	Need to validate within review
	Generative LLMs	Outputs are highly dependent on the prompts the user provides for each data element. To use within a review, careful evaluation is required to consider the types of errors identified (including hallucinations and missing data).	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Need to validate within review
Quantitative extraction	Generative LLMs	Outputs are highly dependent on the prompts the user provides for each data element. To use within a review, careful evaluation is required to consider the types of errors identified (including hallucinations, incorrect vales, and missing data).	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Need to validate within review
Image extraction	Generative LLMs	As part of a dedicated web interface, there are examples of generate AI models being used suggest axes and/or points to extract on a figure, with human correction and verification. For clearly defined figures (e.g. bar plots), this is appropriate and may offer efficiency gains. Humans must retain full oversight over the process to ensure no errors are made. Research on extraction from more complex images is ongoing.	Web Plot Digitizer AI features (https://automeris.io/)	Human verification required
Data synthesis				
Quantitative analysis	Rule-based	Statistical meta-analysis can be performed using dedicated software packages or web applications. Statistical parameters should be clearly documented and flexible for users to amend to suit their analytical choices.	R packages (e.g. Meta, Metafor), CRSRU Meta-analysis apps	Acceptable for use
	Generative LLMs	The use of LLMs to <i>synthesise</i> (as opposed to summarise) results across studies.	Google AI Mode, Elicit, Perplexity, Asta	Not acceptable for use
Code drafting and troubleshooting	Generative LLMs	Generative models can provide solutions to coding problems and/or draft entire scripts based on instructions. Substantive knowledge of the coding language is recommended to ensure that the code performs as intended. All	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Human verification required

		outputs require human checking to ensure there are not unintended errors which alter results.		
Qualitative analysis	Generative LLMs	There is growing interest in using generative models to summarise text, generate codes or subcodes, and identify themes. However, performance will depend heavily on the prompts provided and the task context.	GenAI features within common qualitative tools including NVivo, MAXQDA	Need to validate within review
	Generative LLMs	The use of LLMs to <i>synthesise</i> (as opposed to summarise) results across studies.	Google AI Mode, Elicit, Perplexity, Asta	Not acceptable for use
Visual synthesis	Rule-based software		EPPI Mapper, EviAtlas	Acceptable for use
Certainty of evidence assessment				
Certainty of evidence assessment	Rule-based	Template tools to support the application of established frameworks. Humans still make certainty judgements.	GRADEPro, CINeMA, CERQUAL	Acceptable for use
	Supervised classifiers	Combinations of tools can be put together in a pipeline to conduct GRADE assessments	URSE	Exploratory and supplementary use
Text Generation				
Drafting	Rule-based	Template tools to support structured reporting.	TERA Review Wizard;	Acceptable for use
	Generative LLMs	Generated templates and/or text from provided instructions (i.e. user could supply a pre-defined protocol or review format). This may be appropriate as an early starting point, but it is not recommended to use Generative AI to summarise results across studies or interpret results. There is a significant risk of incorrect conclusions or misinterpretation. All outputs must be refined extensively by human reviewers.	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Exploratory and supplementary use
Revisions	Generative LLMs	Generative AI can be used to improve language in an existing draft e.g. reducing word count, simplifying language, or using alternative phrasing. These tools may also be appropriate for turning bullet points or rough drafts into structured paragraphs.	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Human verification required

Plain language summaries	Generative LLMs	Generative models have demonstrated abilities to make technical language more accessible. While it is not recommended to use Generative AI to summarise results across studies, this may be appropriate for translating a (human drafted) section on the overall review findings into a plain language summary. All outputs require human oversight to ensure the review findings are not misrepresented.	ChatGPT, CoPilot, Claude, Gemini, DeepSeek	Human verification required
Machine translation of non-English studies	Generative and other machine learning approaches	There is some evidence that machine translation can be helpful, but is not sufficiently accurate for more complex tasks.	Google Translate; LLMs	Human verification required
Agentic end-to-end evidence synthesis				
Multiple tasks executed autonomously	Agentic AI	AI agents perform steps which appear similar to the evidence synthesis process, with or without human input/refinement at each stage. This may result in a final report which aims to summarise the findings across included studies. Generative AI is not reliable for summarising results across studies and should not be used for this purpose. Furthermore, given the summative number of errors likely to have arisen across each stage of the process, it is not appropriate to use these tools for evidence synthesis.	Elicit, Perplexity	Not acceptable for use

2: Selecting and using an AI tool

Who is this section for?

- Evidence synthesists
- Trainers of evidence synthesis methods
- Evidence synthesis organisations

There are several issues to consider before deciding to use an AI system in each evidence synthesis project. Clearly, once a team has used a given system before, then some of the issues outlined below will not need to be considered in detail, but there are still important issues that might be summarised as:

Is the use of this AI system justified in my review?

The questions below are intended to aid in answering this question.

Is there a beneficial AI tool which exists for your synthesis task?

As more AI tools become available, there will likely be one or more tasks in each synthesis project amenable to automation. However, many issues are relevant when considering a system for use in a review, a series of reviews, or across an organisation. The primary advantage offered by many AI systems is time savings, but the work involved in gaining system access, team training, and evaluation activities may mean that expected efficiency gains do not materialise. Training and other short-term costs are likely to disincentivise AI for single projects, with lower relative costs and greater benefit over time. Any system you consider should be trialled to assess its fitness for purpose. Other reasons to consider the use of AI systems may include the ability to conduct more complex or larger-scale synthesis, to identify human errors, to increase the accuracy of human input, or to reduce the resources dedicated to certain tasks.

At a minimum, any proposed AI tool needs to align with existing workflows and uphold core personal and organisational principles. Critics contend that LLM-based models pose significant risks for research applications due to inherent biases, potential inaccuracies, and ethical concerns. Additionally, concerns have been raised about substantial environmental impact stemming from the computational resources required for development and operation. Financial support and routine use of these systems may inadvertently cause harm, perpetuating a culture and an economy of exploitation and inequality. Access inequities should also be considered: high costs and infrastructure requirements may exclude researchers in lower-resource settings from using the most effective tools. Conversely, others point to the possible benefits their use may bring, including more efficient evidence-to-decision pipelines, minimising research waste and downstream benefits for people's lives.

Does your team have the resources to use the tool?

Evidence synthesis teams will require resources to use any AI system. This might require financial resources to buy access or technical resources to install and run it locally. Many cutting-edge systems are available only as open-source repositories, such as on GitHub. In these systems, a team

needs considerable technical expertise to build and run the code, knowing that technical support is likely to be hard to come by (i.e., the team needs to be self-sufficient in this regard).

Access to systems can also be restricted by cost. Using an LLM can be very expensive, so companies must charge for access to cover their expenses. Therefore, few freely available LLM-based systems are likely to be fully accessible in the foreseeable future; access to the best features will likely be limited to paying customers.

Assuming a team has access to a system, the next issue to consider is whether they can use it responsibly and appropriately. Depending on the application and sensitivity of the data evaluated (e.g. proprietary or personally identifiable information), data security may be a consideration in selecting a system. Some training may be necessary to enhance understanding of AI systems. Alternatively, you might consider recruiting an author to your team with experience in computer science and AI, much as one would recruit an information specialist for research or a statistician for data analysis.

Teams should also consider time and maintenance demands. Effective integration requires initial setup, validation, and periodic updates to sustain performance. Ongoing use, therefore, depends on dedicated time, personnel, and processes to ensure continued reliability.

Is the tool validated and appropriate for your specific task and context?

According to the PRISMA reporting guidelines, authors should justify their use of an automation system by referencing a relevant prior evaluation.⁵¹ This prior evaluation should address both the specific use of the system and the context of that use; that is, whether the tool has been trained and tested in a manner pertinent to your evidence synthesis task. For example, if a classifier designed to identify randomised controlled trials (see Section 1, Box 2 of Paper 2 on *building and evaluating evidence synthesis tools*) was developed using records from the Embase database, it may perform well on similar records (e.g. the titles and abstracts of reports of research in healthcare). However, if the same classifier is applied to a different type of record, such as those sourced from a clinical trials registry or outside healthcare, the system's performance is likely to decline significantly, rendering it an invalid choice for those contexts. Ultimately, a well-designed tool with high internal validity remains unsuitable if its generalisability to your specific use case is not clearly established.

In the case of generative AI, prompts used with LLMs require careful review before they can be assumed to be reusable across reviews. Not only can the apparently random way results can change pose challenges for evaluation, but the large commercial models themselves frequently change in unpredictable ways, and prompts designed for one LLM may work entirely differently on another or change performance when the LLM version is updated. For this reason, the prompt text and model version should be recorded in sufficient detail to support monitoring and reproducibility. Consider using version control for prompts and maintaining a changelog when prompts are modified.

For many systems, there may be no prior evaluations to justify their use in a given review, either because there are (almost) none or because the evaluations that do exist are too small or not sufficiently similar. Given the relative novelty of LLM-based systems and the inherent uncertainty in how they might be evaluated, it would be sensible to assume that some validation work may be necessary to justify their use in any review at present. Undertaking a 'Study Within A Review' (or

SWAR) could be a reasonable option, as it may allow contributions to the emerging evidence base while also enabling the necessary monitoring of AI system performance.

While using an AI system (implementation)

When rolling out a new system for use across a team, there will be a need for education and training. Once in use, however, all AI systems should be monitored, and their algorithmic outputs audited to ensure they continue to work as expected, particularly when the version is updated. Potential risks to the review process – or its reliability – should be considered at the outset, and appropriate risk management plans should be developed.

After using an AI system in an evidence synthesis

After using an AI system or tool in an evidence synthesis, it is helpful to reflect on lessons learned from its application. Did it perform as expected and address the issues it was designed to solve? Are there implementation insights to consider for future use? This feedback is crucial for developers to understand how to improve their tools.

Teams should document lessons learned in a structured format, including, for example: (a) whether performance matched expectations; (b) any errors or unexpected behaviours; (c) time/resource savings achieved; and (d) recommendations for future use. This documentation should be retained in the review record.

3: Assessing the suitability of an AI tool for use in an evidence synthesis

Who is this section for?

- Evidence synthesists
- Evidence synthesis methodologists
- Organisations producing evidence synthesis
- AI development teams

Background to this section

The following is based on the *Responsible Handover Framework* by Sense About Science.⁵² It also expands on the [CDA-AMC's evaluation instrument](#) and incorporates additional RAISE recommendations for AI system developers and Cochrane's process for [proposing changes to methods or tools](#). Those assessing AI tools may also want to consider other transparency or trustworthiness frameworks, such as those developed by the National Institute of Standards and Technology⁵³, including their AI Risk Management Framework trustworthiness characteristics.

About AI tool assessment

AI tools have been developed for many parts of the evidence synthesis process in numerous different domains. It is important to consider the 'fit' of the tool for an intended use case before beginning to use it in a review. Key issues to address are:

1. What is the purpose of the tool?
2. Where have the training, testing and validation data come from?
3. Is the tool validated and performing sufficiently well for use?
4. Usability and user capability
5. Transparency, licenses, availability & documentation

Before beginning the assessment, teams should agree on decision criteria: what level of evidence or transparency will be required before a tool can be adopted, and what will trigger a decision not to proceed. This helps ensure that the assessment leads to clear, defensible decisions.

"Responsible handover of AI" was developed by Sense About Science in collaboration with its partners.⁵² This is a framework that supports consideration of a given AI tool for a specific task. It covers the information that should be available when an AI tool development team deploys a tool, as well as the questions that evidence synthesis methodologists should ask when developing best-practice guidance or considering tool adoption.

In line with the mutually supportive and dynamic ecosystem envisaged by RAISE, we recommend that methodologists and organisations work with AI development teams to inform the development of tools that is designed to support evidence synthesis effectively. When ready to deploy, AI tool development teams, methodologists (and/or organisations) should jointly address the questions below to inform adoption decisions and best-practice guidance. This framework can also be used when reporting the results of an evaluation that was a joint effort between evidence synthesis methodologists and AI development teams; for example, it could be included as supplementary material to a journal article. (We also recognise that best practice guidance is currently limited, and

that evidence synthesists are also making decisions about which tools to use; this process might currently be applicable for them too.)

Each assessment is specific to function and context; for platforms or tools with multiple embedded AI systems, each AI system requires a separate assessment. (For example, a systematic review platform might contain several AI systems for different tasks such as screening and data extraction.) Without sufficient disclosure from AI developers, this assessment is not possible. In these cases, consider whether the tool can be used only for low-risk, non-decisional support (with full human verification) or whether adoption should be deferred until the missing information is provided.

Ultimate responsibility for a specific evidence synthesis lies with the evidence synthesist, regardless of who completed the Responsible Handover questionnaire. This includes responsibility for verifying outputs, documenting AI use, and ensuring that conclusions remain supported by the underlying evidence.

How to decide on AI tool use after this assessment

Potential answers to each question are provided throughout the framework (see Figure 1). An editable form is available as supplementary material (file 1)³ for completion whenever a tool is made available by AI development teams or considered for adoption by members of the evidence synthesis community. An example of a completed form is available (file 2). These should be used to assess the risk or readiness profile of the AI tool or system. Could its use (or misuse, malfunction, etc.) in an evidence synthesis potentially invalidate the findings, render the conclusions unreliable, introduce bias, or decrease the certainty of the conclusions? Consider whether the risk or the tool's readiness is high, medium, or low, and pairing this with a decision: proceed; proceed with defined mitigations and monitoring; or do not proceed until additional evidence or disclosures are available.

As a general guide: (1) Proceed is appropriate when validation evidence is strong, transparency is high, and risks are well-understood and reasonably manageable. (2) Proceed with mitigations applies when the tool shows promise but has gaps in evidence, requires additional monitoring, or presents moderate risks that can be actively managed. (3) Do not proceed may be warranted when validation evidence is absent or weak, transparency is insufficient for meaningful assessment, or identified risks cannot be adequately mitigated. Teams may consider stopping signals that result in pausing the assessment and reconsider whether they wish to proceed only with additional evidence or disclosures. They could include, but are not limited to:

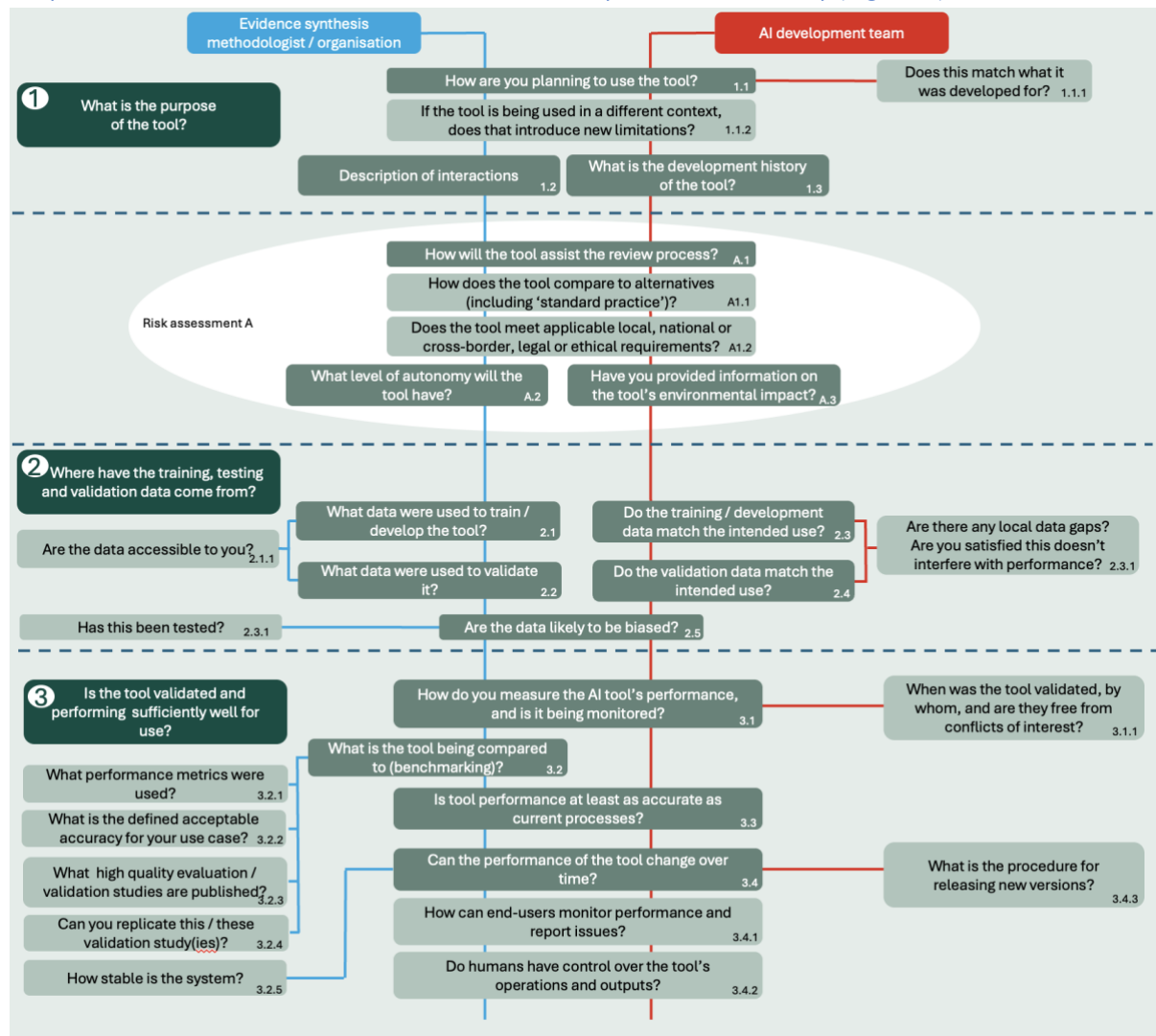
- a. No published validation in a relevant context
- b. Concerns about the replicability of the validation
- c. Concerns about performance claims based only on the developer's own validation and /or methodological limitations
- d. Lack of legal or policy compliance at the evidence synthesist's or tool user's organisation, or at the national or international level
- e. Terms allow use of your content for model training without opt-out

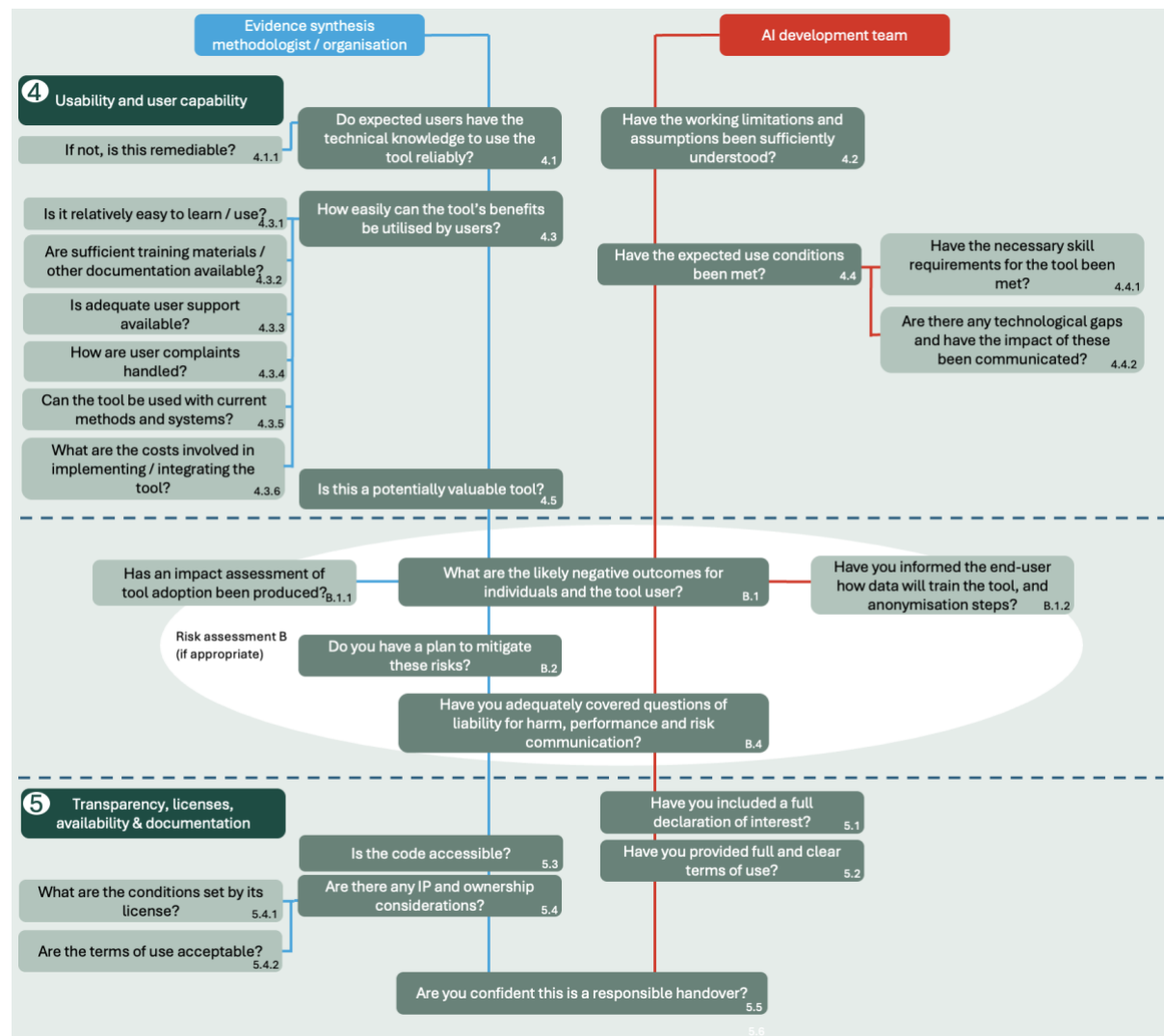
³ At the time of writing in late 2025, teams should be encouraged to complete the form available as a Word document. Sense About Science is planning to release an interactive online tool in due course to support these assessments. Once this is available, the recommendation will be to use this new online tool going forwards.

- f. Inappropriate level of human oversight available, e.g. no way to monitor or audit outputs
- g. The AI tool developer is unresponsive to questions

The framework can be applied at any stage of an AI tool's development and implementation. It can also be repeated at different stages. In practice, it should be revisited whenever the tool, model, prompt set, or deployment context changes. However, given that an important section of the form concerns tool validation, using the framework after some validation is completed would be optimal.

Responsible handover of an AI tool to the evidence synthesis community (Figure 1)





Specific considerations for an evidence synthesist or user of an AI tool

The following summarises specific considerations for an evidence synthesist or user of an AI tool when applying the responsible handbook framework to decide whether to use an AI tool.

What is the purpose of the AI tool?

- ☐ Is the intended use case clearly defined?
- ☐ Does it match your specific task and context?
- ☐ Is the level of human oversight appropriate?
- ☐ For LLMs, is the model version and prompt recorded?

Tool name, version and developer name (and country)**Source URL and how to access**

The AI system must be accessible, either via a live platform, beta version or equivalent access method.

Description of the interactions

Do you understand the task the AI system claims to perform, as well as the strengths and limitations? Strengths and limitations relate to the domains and breadth of the data used for training, testing and validation (see sections below) and should include what scenarios the AI tool is best suited to and when it may be inappropriate to use. If the AI tool uses generative AI or large language models (LLM), identify which LLM is used. Understand how the tool should be used, including whether your proposal is to replace an existing task or method or if it is to be used in parallel with current methods. If the latter, understand which existing task or method this new approach should complement or be used alongside. Ensure there are details about how human oversight is ensured within the AI tool, confirming how it is set up so that humans have control of its operations and outputs.

How are you planning to use the tool

Consider the risks and the benefits of using the tool, including how the tool will assist in the review process, whether it is clearly better than alternatives (whether they be other AI tools or manual processes) and what level of autonomy the tool will have (how has human oversight been ensured and do humans have control of the AI tools operations and outputs).

Where have the training and testing data come from?

- ☐ Are training and testing data sources disclosed and accessible?
- ☐ Do training and testing data match your evidence synthesis domain/context?
- ☐ Are there known limitations or biases (language, geography, date range, methods, etc.)?
- ☐ Is there a separation between the training, testing and validation datasets?

Determine whether the data used for testing (and training, where appropriate) the tool match your synthesis context. Consider the strengths and limitations of the data in relation to the domains and breadth of the training and testing data used, including whether it aligns with your review topic and scope, or whether any biases or equity issues might undermine your review findings or conclusions. Consider the task the AI system claims to perform and whether the context, strengths and limitations of the AI tool apply to the context of your synthesis, including

whether the sources (e.g. open access only, PubMed only), geography, language, methods, etc., would have a detrimental impact on your evidence synthesis. Consider:

- *Are the sources of training and validation data clear and transparent?*
- *Is the corpus large enough for your work (is it adequately sized and not skewed towards certain topics)?*
- *Does it contain the breadth of topics applicable? Are there gaps in the scope and domain, sources, geography, language, methods, publication types, etc.?*
- *Is the age of the data adequate for your needs? Is the corpus frequently updated? Are there gaps for older and newer information?*
- *Whether there is any risk of contamination between datasets that were used for training, testing and validation (as they could affect performance metrics)?*

Is the AI tool validated and perform sufficiently for use?

- ☐ **Has performance been validated?**
- ☐ **Is it reported in sufficient enough detail to be transparent and replicable?**
- ☐ **Are metrics appropriate and does performance meet your acceptable threshold?**
- ☐ **Is the validation published and / or peer-reviewed?**
- ☐ **Can performance change over time?**

Consider:

- *Date(s) of the validation and whether it relates to the version you will be using.*
- *Whether the evaluations and / or validation studies are publicly available and whether they are published in a peer reviewed journal (i.e., were there any quality assurance processes)*
- *Whether they were conducted by the AI developers or independent methodologists (the latter of which would likely not have a conflict of interest in the AI's performance).*
- *Whether the methods are reported in sufficient enough detail to be transparent and replicable.*
- *Whether the performance metrics were appropriate to address the research question? RAISE 2 includes a taxonomy of performance metrics that should be considered. Depending on the performance metrics, does the evaluation include an appropriate comparator (e.g. current best practice, human as gold standard, another AI system that has previously been evaluated against current best practice)? Some metrics, such as user experience, may not need a comparator.*
- *Whether there has been any risk of contamination between the training, testing and validation datasets.*
- *Whether the evaluation, including methodological rigor and tone of reporting, supports the performance claims.*
- *Whether the evaluation is transparently and comprehensively reported to enable replication of the study?*

Determine whether tool performance is at least as accurate as current processes, or accurate enough against expected benchmarks (depends on the AI system).

Can the performance of the tool change over time?

Understand how you as a user of the AI tool can monitor performance and report issues, especially if performance of the tool could change over time, e.g. it depends on third party language models that can be updated independently. Also understand how the AI development team will share information about new version releases of new versions and how changes in performance will be communicated and managed.

Usability and user capability

- ☐ Does your team have the skills to use it reliably?
- ☐ Is training/documentation available?
- ☐ What are the costs (financial, time, infrastructure)?
- ☐ Is user support available?

Do you have the technical knowledge to use the tool reliably; and if not, is this remediable, e.g. can you recruit someone with this expertise to your review team? Consider how easily can the tool's benefits be utilised? In particular:

- *Is it relatively easy to learn / use?*
- *Are sufficient training materials and other documentation available, and are they user friendly?*
- *Is adequate user support available?*
- *Are the costs to use the AI tool acceptable?*
- *Are we aware of user complaints and how they were handled?*
- *Can the tool be used with current platforms and systems?*

Consider carefully likely negative outcomes if the tool is used, and how you plan to mitigate any risks identified.

Transparency, licenses, availability and documentation

- ☐ Are the terms of use clear and acceptable?
- ☐ Will uploaded content be used for training? Can you opt out?
- ☐ Does the tool meet your legal or ethical requirements?
- ☐ Is there a conflict-of-interest disclosure?

Are the terms of use acceptable? This may require formal legal advice. We particularly encourage you to consider whether the system has data security procedures appropriate for intended use, for example:

- *Will it claim ownership of uploaded content or impose limitations on its use.*
- *Will it gain rights to use the uploaded content for distribution, repurposing, or adaptation to train the AI system.*
- *Will it impose any limitations on the use of the AI system's outputs*
- *Does it have privacy features that allow users to disable data collection for training and turn off or delete history or interaction tracking.*

Assess issues concerning the usability and stability of the system. This includes, for example:

- *Does the tool meet other local, national or cross-border legal or ethical requirements, e.g. data privacy, IP, copyright, anti-discrimination etc., and are the potential societal and environmental impacts acceptable? This may require formal legal advice.*

- *Is the system run on external servers, or can it be run locally?*
- *Whether the AI tool is available for you to use where you are based and in an appropriate language.*
- *Does the system have a track record or plan to suggest it will be available and maintained?*

Understand the funding sources of the AI tool and consider whether there are any conflicts of interest

Consider whether the AI tool is owned by a commercial organization that can gain financially from use of the tool; financial gain could include selling it, licensing it, making it available through paid subscription, etc. Sources of funding should ideally be declared by the AI tool.

4. Ethical, legal and regulatory considerations when using an AI tool

Who is this section for?

- Evidence synthesists
- Evidence synthesis methodologists
- AI tool development teams
- Organisations that produce evidence synthesis
- Users of evidence synthesis
- Trainers of evidence synthesis methods

As the user of the AI system, it is your responsibility to ensure that you do not breach the following guidelines. Publishers, including STM and Wiley, and research integrity organisations, including UKRIO⁵⁴ are developing helpful guidance to help navigate these issues when using AI tools^{45,55}, which are summarised below along with considerations and tips on how to navigate the potential risks. Three areas warrant attention beyond the table. First, evidence synthesis teams may process material subject to data protection law, including unpublished manuscripts, patient data, and confidential communications. Before using cloud-based AI tools, teams should confirm whether their institution's data processing agreements cover the provider, whether data transmission complies with any confidentiality obligations, and whether the provider's terms of service permit the use of submitted content for model training. In the EU and EEA, processing personal data through an AI system will typically require a lawful basis under GDPR and may require a data protection impact assessment. Second, legal liability for AI-assisted evidence synthesis errors is untested in most jurisdictions, but using an AI tool does not transfer responsibility for accuracy from the review team to the tool provider. Teams should document their validation steps and human oversight procedures to demonstrate responsible use. Third, uploading published text, figures, or tables to an AI system may constitute reproduction of copyrighted material. Teams should not assume that access for reading implies permission for AI processing and should explicitly check the publisher's terms of service.

Potential risks of using AI systems*	Considerations and how to mitigate risks
Plagiarism and provenance How to ensure you do not present the work of others (words, data or ideas) as your own without appropriate acknowledgement.	Some AI systems may misrepresent the provenance and attribution of research. This is an area of active development as it is important that you are not inappropriately claiming the ideas of others or inappropriately attributing them. The EU Artificial Intelligence Act (see https://artificialintelligenceact.eu/high-level-summary/) recognises the need for transparency around authorship of any content from AI systems. <i>Tips:</i> - <i>You are ultimately responsible for the accuracy of your work and so it is important to verify the AI system's outputs, including claims, citations, etc.</i> - <i>Select AI systems that have clear and transparent information about processes and methods. Ideally there should be public information about how it works and its limitations; a history of development, including how</i>

	<i>identified errors or biases are addressed; and how it continuously monitors and improves the model.</i> - <i>Where relevant, ensure AI system outputs include the correct references and that you cite them properly in your work.</i>
Copyright and intellectual property (IP) With copyright and IP, you should be wary of two situations: (1) ensuring you do not breach copyright, license agreements or infringe third-party intellectual property reuse when uploading content to AI systems, and (2) protecting the IP of your own work when using AI systems (not only is this important for your own work, but if you unintentionally assign rights to your work to the AI system, it could compromise your ability to publish the work at a later date).	Authors make decisions about how their research should be used by publishing under different licenses. These licenses define the copyright and reuse requirements (see https://www.copyright.com/education-copyright-foundations/ and https://cla.co.uk/). Many popular permissive licenses require attribution for reuse, and many LLMs are not currently able to do this properly, though it is an active area of development (for example, Creative Commons is exploring preference signals (https://creativecommons.org/2024/08/23/six-insights-on-preference-signals-for-ai-training/)). There are ongoing developments with numerous lawsuits from content creators whose copyrighted materials have been used to train LLMs and similar technologies, though AI developers typically claim the content was used under fair use principles (Baker & Hostetler’s tracker includes some examples of ongoing copyright litigations: https://www.bakerlaw.com/services/artificial-intelligence-ai/case-tracker-artificial-intelligence-copyrights-and-class-actions/). Many publishers are clarifying how their content can be used in AI tool development, for example, by updating copyright lines to be explicit (e.g. allowing the content to be used for training but not for the outputs unless they have permission from the license holder) or evolving model reuse licenses for AI content (e.g. template licenses covering essential legal terms and conditions). This is an active area of development. Most AI-generated outputs are not protected by copyright. Unless otherwise specified in the terms and conditions, and especially for publicly available LLM-based AI tools, users should assume there are limited controls over the future use of content uploaded to the platform (e.g. for logging, product improvement or model training. A generative LLM may (re)produce copyrighted material – and may also reproduce personal or sensitive information – so outputs from these systems should always be checked. <i>Tips:</i> - <i>Ensure adherence to applicable laws depending on where you are based – for example, US copyright law is not the same as UK copyright law.</i> - <i>Where possible, use AI tools that have been specifically developed on curated, trusted content (as opposed to publicly available platforms).</i> - <i>Check for privacy features in AI tools; ideally, use systems with options to disable data collection for</i>

	<i>training and turn off or delete history or interaction tracking. Always double check default settings before using the system.</i> - <i>Once you upload content into an AI system, you will not be able to easily remove it, if at all. Ensure you are clear on any copyright or third-party IP considerations before using an AI system, ensure you own or have the relevant rights to upload content into an AI system, and ensure that you have the rights to reuse the AI system outputs in your own work (check the terms and conditions for restrictions on its use, e.g. to share, adapt, modify, or any sublicensing rights).</i> - <i>When assessing the terms and conditions, pay particular attention to clauses on data ownership, data handling and usage rights, including references to whether the tool trains on or retains your content; it is strongly recommended you do not use AI systems that gain any rights to your content, including the right to use, distribute, repurpose or adapt your content to train the AI system.</i> - <i>Look for AI tools with strong privacy controls and clear terms and conditions. Some organisations or institutions may have access to private or enterprise versions of AI systems that offer greater privacy and security features, with clearer terms and conditions than free and public AI systems. For example, the pro-version of Claude guarantees that input will not be used for LLM training, whereas the free version does not; therefore, for example, uploading copyrighted PDFs to the free-version could be a copyright infringement. Each version needs assessing before use to ensure they align with the publisher policies.</i> - <i>If in doubt, seek legal advice.</i> <i>When considering IP protection terms and conditions, Wiley has useful advice about terms and conditions ⁵⁵:</i> - <i>Be wary of: “perpetual” rights to your content; “royalty-free” usage rights; “transferable” rights to third parties; broad statements about using content “for training” or “for any purpose”; classifying the AI-generated content as “open”, “free to use”, or subject to a permissive license or “Creative Commons” license (which may open use of your work up to the rest of the world), or subject to a restrictive license; any limitations on your ability to use your content or the output; or unclear policies about data retention and deletion.”</i> - <i>Instead, look for ones that: “explicitly state they do not claim rights to your content; put no limitations on your use of your content or the output; have clear data deletion policies; specify limited uses for user content; provide transparent information about data retention; and allow you to control or opt out of data training.”</i>
--	--

Jurisdiction and licensing	Some legislation prohibits the ‘export’ of certain services in some parts of the world, for example, the use of a USA-based LLM systems in other countries. In addition, access to systems can be limited while they are rolled out, due to availability of the necessary compute resources. This means that the newest technologies are sometimes only available in some regions, such as only the USA itself. <i>Tips:</i> - <i>Ensure you have a license agreement to use the AI system.</i> - <i>Ensure you are clear on any restrictions on its use or licensing considerations before using an AI system</i> - <i>If in doubt, seek legal advice.</i> - <i>Ensure you are complying with your organisation’s policy on AI use.</i>
Terms and conditions of use	<i>Tips:</i> - <i>See ‘copyright and IP’ row above for relevant terms and condition tips</i> - <i>Ensure you understand the terms and conditions for using the AI system before use.</i> - <i>Check the terms and conditions allow you to use the AI system in the way you want to and ensure any iterations are read to ensure continued compliance.</i> - <i>Consider undertaking training on the AI system before using it with your own content.</i> - <i>Check with your employer that using the AI system will not contravene their insurance.</i> - <i>If in doubt, seek legal advice.</i>
Ethical considerations How to ensure you do not breach conditions of informed consent or risk harm or discrimination of individuals (e.g. lack of information privacy, protection and security, as well as being aware of algorithmic bias)	<i>Tips:</i> - <i>Ensure your use of an AI system aligns with confidentiality, compliance and privacy responsibilities, including data protection laws.</i> - <i>Be cautious about whether you can upload sensitive content, such as private or personal information, content under confidentiality agreements or embargoes (e.g. unpublished work), data or content protected by regulations or IP and copyrighted (see ‘copyright and IP’ row above). Consider removing this before uploading content to AI systems.</i> Algorithmic bias can perpetuate and even amplify existing societal inequalities if not carefully addressed. Be mindful of the ways that large technology companies have used ‘reinforcement learning through human feedback’ to minimise ‘toxic’ or sensitive output from LLMs. To do this, people have

	been exposed to potentially very extreme content without proper psychological help.
Considerations regarding impact on environment and society	AI has considerable environmental impacts and leads to significant environmental costs throughout its entire life cycle – from the extraction of raw materials (including minerals), building supporting infrastructure, to the energy-intensive phases of model training and deployment, followed by disposal.⁵⁶ Specifically, AI consumes large amounts of energy (see ^{57,58}) and water, particularly in data centres where cooling and continuous operation are required. The use of minerals, energy and water comes with carbon emissions. ^{56,59} It is increasingly recognised that these factors, among others can also have significant societal impacts on local communities and access to amenities. High costs, infrastructure requirements, and jurisdictional limits can restrict access for researchers in low-resource settings, potentially widening global inequities in who can perform advanced evidence synthesis. <i>Tips:</i> - <i>Think carefully about the environmental impact of AI tools in your selection process and prioritise those with transparent reporting of, and efforts to minimise, their carbon footprint.</i> - <i>Balance the environmental cost against potential benefits, such as reduced research waste through more efficient synthesis. Select AI solutions that use renewable energy for their data centres, where this is transparently documented.⁶⁰</i> - <i><u>The Green Algorithms project (https://www.green-algorithms.org/) groups calculators and resources that researchers can use to estimate and reduce the carbon footprint of their project.</u></i> For institutional users, include environmental impact in procurement assessments alongside performance, cost and legal considerations. - Encourage consideration of accessible, open, or institutionally supported tools; acknowledge equity implications in policy and procurement decisions.

References

1. O'Connor AM, Clark J, Thomas J, et al. Large language models, updates, and evaluation of automation tools for systematic reviews: a summary of significant discussions at the eighth meeting of the International Collaboration for the Automation of Systematic Reviews (ICASR). *Systematic Reviews*. 2024;13(1):290. doi:10.1186/s13643-024-02666-2
2. Doneva SE, Qin S, Sick B, et al. Large language models to process, analyze, and synthesize biomedical texts: a scoping review. *Discover Artificial Intelligence*. 2024;4(1):107. doi:10.1007/s44163-024-00197-2
3. Adam GP, Davies MC, George J, et al. Machine Learning Tools To (Semi-)Automate Evidence Synthesis: A Rapid Review and Evidence Map. Published online 2024. doi:10.23970/AHRQEPWHITEPAPERMACHINE
4. Clark J, Barton B, Albarqouni L, et al. Generative artificial intelligence use in evidence synthesis: A systematic review. *Res synth methods*. 2025;16(4):601-619. doi:10.1017/rsm.2025.16
5. Lieberum JL, Toews M, Metzendorf MI, et al. Large language models for conducting systematic reviews: on the rise, but not yet ready for use—a scoping review. *Journal of Clinical Epidemiology*. 2025;181:111746. doi:10.1016/j.jclinepi.2025.111746
6. Kataoka Y, So R, Banno M, et al. Large language models for automated PRISMA 2020 adherence checking. *arXiv*. Preprint posted online 2025. doi:10.48550/ARXIV.2511.16707
7. Gupta R, Park JB, Bisht C, et al. Expanding Cosmetic Plastic Surgery Research With ChatGPT. *Aesthetic Surgery Journal*. 2023;43(8):930-937. doi:10.1093/asj/sjad069
8. Paynter RA, Featherstone R, Stoeger E, Fiordalisi C, Voisin C, Adam GP. A prospective comparison of evidence synthesis search strategies developed with and without text-mining tools. *Journal of Clinical Epidemiology*. 2021;139:350-360. doi:10.1016/j.jclinepi.2021.03.013
9. Adam GP, DeYoung J, Paul A, et al. Literature search sandbox: a large language model that generates search queries for systematic reviews. *JAMIA Open*. 2024;7(3):ooae098. doi:10.1093/jamiaopen/ooae098
10. Lau O, Golder S. Comparison of Elicit AI and Traditional Literature Searching in Evidence Syntheses Using Four Case Studies. *Cochrane Evidence Synthesis and Methods*. 2025;3(6):e70050. doi:10.1002/cesm.70050
11. McKeown S, Mir ZM. Considerations for conducting systematic reviews: A follow-up study to evaluate the performance of various automated methods for reference de-duplication. *Research Synthesis Methods*. 2024;15(6):896-904. doi:10.1002/jrsm.1736
12. Guimarães NS, Ferreira AJF, Ribeiro Silva RDC, et al. Deduplicating records in systematic reviews: there are free, accurate automated ways to do so. *Journal of Clinical Epidemiology*. 2022;152:110-115. doi:10.1016/j.jclinepi.2022.10.009
13. Affengruber L, van der Maten MM, Spiero I, et al. An exploration of available methods and tools to improve the efficiency of systematic review production: a scoping review. *BMC Medical Research Methodology*. 2024;24(1):210. doi:10.1186/s12874-024-02320-4

14. Effective Health Care Program A for HR and Q. *Machine Learning Tools To (Semi-)Automate Evidence Synthesis: A Rapid Review and Evidence Map*. Agency for Healthcare Research and Quality; 2024. [3]
15. Ge L, Agrawal R, Singer M, et al. Leveraging artificial intelligence to enhance systematic reviews in health research: advanced tools and challenges. *Systematic Reviews*. 2024;13(1):269. doi:10.1186/s13643-024-02682-2
16. Callaghan M, Müller-Hansen F, Bond M, et al. Computer-assisted screening in systematic evidence synthesis requires robust and well-evaluated stopping criteria. *Systematic Reviews*. 2024;13(1):284. doi:10.1186/s13643-024-02699-7
17. Repke T, Tinsdeall F, Danilenko D, et al. Don't Stop Me Now, 'Cause I'm Having a Good Time Screening: Evaluation of Stopping Methods for Safe Use of Priority Screening in Systematic Reviews. *Cochrane Evidence Synthesis and Methods*. 2026;4(1):e70068. doi:10.1002/cesm.70068
18. Callaghan MW, Müller-Hansen F. Statistical stopping criteria for automated screening in systematic reviews. *Syst Rev*. 2020;9(1):273. doi:10.1186/s13643-020-01521-4
19. Oami T, Okada Y, Nakada TA. Performance of a Large Language Model in Screening Citations. *JAMA Network Open*. 2024;7(7):e2420496. doi:10.1001/jamanetworkopen.2024.20496
20. Huotala A, Kuuttila M, Ralph P, Mäntylä M. The Promise and Challenges of Using LLMs to Accelerate the Screening Process of Systematic Reviews. Published online 2024. <https://arxiv.org/abs/2404.15667>
21. Homiar A, Thomas J, Ostinelli EG, et al. Development and evaluation of prompts for a large language model to screen titles and abstracts in a living systematic review. *BMJ Ment Health*. 2025;28(1):e301762. doi:10.1136/bmjment-2025-301762
22. Shemilt I, Thomas J, Churchill R, et al. Generative artificial intelligence tools versus conventional screening by humans for selecting eligible study reports for evidence synthesis: a living study within a review (SWAR). Published online 2025. doi:10.17605/OSF.IO/G7MKB
23. Lee DCW, O'Brien KM, Presseau J, et al. Accuracy and efficiency of using artificial intelligence for data extraction in systematic reviews. A noninferiority study within reviews. *Public and Global Health*. Preprint posted online February 27, 2026. doi:10.64898/2026.02.25.26347053
24. Hair K, Wilson E, Maksym O, Macleod MR, Sena ES. A Systematic Online Living Evidence Summary of experimental Alzheimer's disease research. *Journal of Neuroscience Methods*. 2024;409:110209. doi:10.1016/j.jneumeth.2024.110209
25. Lee J, Yoon W, Kim S, et al. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Wren J, ed. *Bioinformatics*. 2020;36(4):1234-1240. doi:10.1093/bioinformatics/btz682
26. Beltagy I, Lo K, Cohan A. SciBERT: A Pretrained Language Model for Scientific Text. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics; 2019:3613-3618. doi:10.18653/v1/D19-1371

27. Pu Y, Beck D, Verspoor K. Enriched knowledge representation in biological fields: a case study of literature-based discovery in Alzheimer's disease. *J Biomed Semant.* 2025;16(1):3. doi:10.1186/s13326-025-00328-3
28. Dhrangadhariya A, Müller H. Not so weak PICO: leveraging weak supervision for participants, interventions, and outcomes recognition for systematic review automation. *JAMIA Open.* 2023;6(1):ooac107. doi:10.1093/jamiaopen/ooac107
29. Devane D, Burke NN, Treweek S, et al. Study within a review (SWAR). *J Evidence Based Medicine.* 2022;15(4):328-332. doi:10.1111/jebm.12505
30. Gartlehner G, Kahwati L, Nussbaumer-Streit B, et al. From promise to practice: challenges and pitfalls in the evaluation of large language models for data extraction in evidence synthesis. *BMJ Evidence-Based Medicine.* Published online December 19, 2024. doi:10.1136/bmjebm-2024-113199
31. Zurrer WE, Cannon AE, Ewing E, Rosso M, Reich DS, Ineichen BV. Auto-STEED: A data mining tool for automated extraction of experimental parameters and risk of bias items from *in vivo* publications. *Neuroscience.* Preprint posted online February 27, 2023. doi:10.1101/2023.02.24.529867
32. Bahor Z, Liao J, Macleod MR, et al. Risk of bias reporting in the recent animal focal cerebral ischaemia literature. *Clinical Science.* 2017;131(20):2525-2532. doi:10.1042/CS20160722
33. Jardim PSJ, Rose CJ, Ames HM, Echavez JFM, Van De Velde S, Muller AE. Automating risk of bias assessment in systematic reviews: a real-time mixed methods comparison of human researchers to a machine learning system. *BMC Med Res Methodol.* 2022;22(1):167. doi:10.1186/s12874-022-01649-y
34. Eckmann P, Barnett A, Bannach-Brown A, et al. Use as directed? A comparison of software tools intended to check rigor and transparency of published work. Wu J, ed. *PLoS One.* 2026;21(2):e0342225. doi:10.1371/journal.pone.0342225
35. Eisele-Metzger A, Lieberum JL, Toews M, et al. Exploring the potential of Claude 2 for risk of bias assessment: Using a large language model to assess randomized controlled trials with RoB 2. *medRxiv.* Published online January 1, 2024:2024.07.16.24310483. doi:10.1101/2024.07.16.24310483
36. Lai H, Ge L, Sun M, et al. Assessing the Risk of Bias in Randomized Clinical Trials With Large Language Models. *JAMA Netw Open.* 2024;7(5):e2412687. doi:10.1001/jamanetworkopen.2024.12687
37. Lai H, Liu J, Bai C, et al. Language models for data extraction and risk of bias assessment in complementary medicine. *npj Digital Medicine.* 2025;8(1):74. doi:10.1038/s41746-025-01457-w
38. Scotti KL, Young S, Gainey MA, Lan H. Artificial Intelligence and Automation in Evidence Synthesis: An Investigation of Methods Employed in Cochrane, Campbell Collaboration, and Environmental Evidence Reviews. *Cochrane Evidence Synthesis and Methods.* 2025;3(5):e70046. doi:10.1002/cesm.70046
39. Owen RK, Bradbury N, Xin Y, Cooper N, Sutton A. MetaInsight: An interactive web-based tool for analyzing, interrogating, and visualizing network meta-analyses using R-shiny and netmeta. *Research Synthesis Methods.* 2019;10(4):569-581. doi:10.1002/jrsm.1373

40. Picot C, Ajiji P, Jurek L, et al. Risk of drug use during pregnancy: master protocol for living systematic reviews and meta-analyses performed in the metaPreg project. *Syst Rev*. 2023;12(1):101. doi:10.1186/s13643-023-02256-8
41. Reason T, Benbow E, Langham J, Gimblett A, Klijn SL, Malcolm B. Artificial Intelligence to Automate Network Meta-Analyses: Four Case Studies to Evaluate the Potential Application of Large Language Models. *PharmacoEconomics Open*. 2024;8(2):205-220. doi:10.1007/s41669-024-00476-9
42. Sousa-Pinto B, Marques-Cruz M, Neumann I, et al. Guidelines International Network: Principles for Use of Artificial Intelligence in the Health Guideline Enterprise. *Ann Intern Med*. 2025;178(3):408-415. doi:10.7326/ANNALS-24-02338
43. Šuster S, Baldwin T, Verspoor K. Analysis of predictive performance and reliability of classifiers for quality assessment of medical evidence revealed important variation by medical area. *Journal of Clinical Epidemiology*. 2023;159:58-69. doi:10.1016/j.jclinepi.2023.04.006
44. Oliveira Dos Santos A, Belo VS, Mota Machado T, Silva ESD. Toward automating GRADE classification: a proof-of-concept evaluation of an artificial intelligence-based tool for semiautomated evidence quality rating in systematic reviews. *BMJ EBM*. 2026;31(1):36-45. doi:10.1136/bmjebm-2024-113123
45. STM Association. *STM Generative AI Paper 2023*. STM Association; 2023. <https://stm-assoc.org/document/stm-generative-ai-paper-2023/>
46. Devane D, Pope J, Byrne P, et al. Comparison of AI-assisted and human-generated plain language summaries for Cochrane reviews: a randomised non-inferiority trial (HIET-1) [Registered Report - stage II]. *Journal of Clinical Epidemiology*. 2026;191:112102. doi:10.1016/j.jclinepi.2025.112102
47. Cadiente A, Implicito C, Udaiyar A, et al. Evaluating Incontinence Abstracts: Artificial Intelligence-Generated Versus Cochrane Review. *Urogynecology*. Published online 2025. https://journals.lww.com/fpmrs/fulltext/9900/evaluating_incontinence_abstracts__artificial.377.aspx
48. Jackson JL, Kuriyama A. How Often Do Systematic Reviews Exclude Articles Not Published in English? *J GEN INTERN MED*. 2019;34(8):1388-1389. doi:10.1007/s11606-019-04976-x
49. Balk EM, Chung M, Chen ML, Chang LKW, Trikalinos TA. Data extraction from machine-translated versus original language randomized trial reports: a comparative study. *Syst Rev*. 2013;2(1):97. doi:10.1186/2046-4053-2-97
50. Jackson JL, Kuriyama A, Anton A, et al. The Accuracy of Google Translate for Abstracting Data From Non-English-Language Trials for Systematic Reviews. *Ann Intern Med*. 2019;171(9):677-679. doi:10.7326/M19-0891
51. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. Published online March 29, 2021:n71. doi:10.1136/bmj.n71
52. Sense about Science. Responsible Handover of AI. 2025. <https://senseaboutscience.org/responsible-handover-of-ai/>
53. National Institute of Standards and Technology (NIST). AI Standards. 2025. <https://www.nist.gov/artificial-intelligence/ai-standards>

54. UK Research Integrity Office. Embracing AI with integrity. Published online 2025.
<https://ukrio.org/wp-content/uploads/Embracing-AI-with-integrity.pdf>
55. Wiley. AI Guidelines for Book Authors. 2025. <https://www.wiley.com/en-gb/publish/book/ai-guidelines>
56. Simon Clark. References for “Should I feel guilty using AI?” 2025.
<https://www.simonoxfphys.com/blog/2025/3/31/references-for-should-i-feel-guilty-using-ai>
57. International Energy Agency (IEA). AI and Climate Change – Energy and AI. 2025.
<https://www.iea.org/reports/energy-and-ai/ai-and-climate-change>
58. Patterson D, Gonzalez J, Le Q, et al. Carbon Emissions and Large Neural Network Training. Published online 2021. doi:10.48550/arXiv.2104.10350
59. Luccioni S, Trevelin B, Mitchell M. The Environmental Impacts of AI – Policy Primer. Published online 2024. doi:10.57967/hf/3004
60. Bergman AC Sarah Hsu, Sara. *Building Green Software*. 2024. Accessed November 7, 2025.
<https://www.oreilly.com/library/view/building-green-software/9781098150617/>
61. Thomas J, McDonald S, Noel-Storr A, et al. Machine learning reduced workload with minimal risk of missing studies: development and evaluation of a randomized controlled trial classifier for Cochrane Reviews. *Journal of Clinical Epidemiology*. 2021;133:140-151.
doi:10.1016/j.jclinepi.2020.11.003
62. Moberg K, Gornitzki C. Combining search filters for randomized controlled trials with the Cochrane RCT Classifier in Covidence: a methodological validation study. *Research Synthesis Methods*. Published online August 28, 2025:1-8. doi:10.1017/rsm.2025.10023

Supplementary file 1: responsible handover tool

What type of handover is this?
Who is completing this form? For example, is it a tool development team handing a tool over to the evidence synthesis community for their use? Is it an evidence synthesis methodologist reporting the results of an evaluation and recommending a tool for use in a specific way? Or is it the result of collaboration between a tool development team and an evidence synthesis methodologist, who are jointly reporting their understanding about the suitability of a tool for use? Record the tool name and version (including model version, where relevant) and the date this form was completed. If any question cannot be answered, record this explicitly (for example, 'Unknown' or 'Not disclosed') and treat missing information as a potential risk signal. Schedule a reassessment date (e.g. annually or when major version updates occur) and record this in the form.
(1) What is the purpose of the tool? Methodologists and AI development teams: agree a statement that defines how the tool should be used (1.1) in an evidence synthesis project. . Record the tool name, version, and deployment context (platform, local installation, API, etc.). If the AI tool uses generative AI, state the model provider and model version, and (where relevant) the prompt set or template version. Then, jointly consider whether there are contextual limitations (1.1.2), informed by input from the AI development team, and whether the intended context of use matches that originally envisaged (1.1.1). State whether the use case is to replace an existing task or method, or if it is to be used in parallel with current methods. If the latter, state which existing task or method this new approach should complement or be used alongside, and what human is required.
(1.2) Methodologists: what interactions have you had with the tool development team? Describe the level of collaboration and any support in its development, if any.
(1.3) AI development teams: describe the development history of the tool briefly. This should include methods or approaches used to safeguard accuracy in the output, a history of data sources, errors discovered, and modifications made. Indicate whether there are details about how the model has developed over time and what future updates are planned.
(Risk assessment A): taking into account the answers given to the previous questions, consider the benefits and risks of using the tool. Evidence synthesis methodologists should consider how the tool will assist in the review process (A.1), whether it is clearly better than alternatives (A.1.1) (for example, whether it outperforms other AI tools or manual processes), and what trade-offs are

introduced (time, cost, bias, transparency, sustainability). Also detail how the tool meets applicable local, national or cross-border, legal and ethical requirements (A.1.3). For example, confirm that the tool does not retain copyrighted material for model improvement, as evidence synthesis teams are rarely granted rights to permit this when downloading papers from publishers. Assign an overall risk level (low/medium/high) and record key mitigations required for use.

Specify the **level of autonomy** (A.2) that the tool should have, including which decisions it is permitted to make without human sign-off, what human oversight is required and how users can override or audit outputs (including an audit trail).

AI development teams should provide information on the tool's **environmental impact** (A.3) and their efforts to minimise its carbon footprint (for example, energy use, model size, data centre information, and any efficiency measures). Where estimates are not available, record this and outline plans to measure and report it.

(2) Where have the training, testing and validation data come from?

(2.1 & 2.2) Methodologists: state **what data were used to train / test / validate the tool**. Specify how they can be accessed and consider whether training dataset was adequately sized and sufficiently broad to cover the stated topics and domains adequately. **Are the data available to you?** (2.1.1 / 2.2.1) Also specify any licensing, permissions or access constraints that affect whether training and validation data can be shared or reused.

Consider:

- Are the sources of training and validation data clear and transparent (including rights and permissions for reuse)?
- Is the currency of the data adequate for your needs?
- Is the age of the data adequate for your needs?
- How data were labelled or annotated as part of building the model (if any)

Determine:

- Power: Is the corpus large enough for your work (is it adequately sized and not skewed towards certain topics)?
- Comprehensiveness: Does it contain the breadth of topics applicable? Are there gaps in the scope and domain, sources, geography, language, methods, publication types, etc.?
- Coverage / currency: Is the corpus frequently updated? Are there gaps for older and newer information?

(2.3 & 2.4) AI tool developer: state whether the data used for training, testing and validation (where appropriate) **match the use case**. Do the training / development data match the intended use? Consider the strengths and limitations of the data in relation to the domains and breadth of the data used (2.3.1), including what scenarios it is best suited to and when it may be inappropriate to use, including scope and domains, sources (e.g. open access only, PubMed only), geography, language, methods, etc. Confirm that there is no contamination between datasets that were used for training, testing and validation.

(2.5) Both: state whether the data are likely to be biased. Consider what biases or equity issues there may have been in the system's training data and strategies for users to help mitigate potential issues due to biases, e.g. were data obtained from published sources only, or open sources only or skewed towards certain topics. Additionally, state whether the corpus was sufficiently large. Describe what steps were taken to ensure training data are representative of intended use case.
(3) Is the AI tool validated and performing sufficiently well for use?
(3.1) Methodologists and development teams: how do you measure the AI tool's performance, and is it being monitored? State the method(s) / protocol(s) used (include URL or attach) and confirm: • <i>Performance metrics measured and results for each</i> • <i>What exclusion criteria were applied</i> • <i>Information about the types and categories of data and detail any potential biases</i> • <i>Any processes used to clean the data, including handling missing data</i> • <i>For generative AI and large language models, state how prompts were developed and evaluated, ensuring the full prompt text is recorded in detail. Also record the model provider and version, key generation settings (e.g. temperature), and any retrieval or tool-use configuration.</i> State who conducted the validation(s), when, and whether they were independent of the AI tool developers. (3.1.1)
(3.2) Methodologists, state the comparator (if relevant), and how this is able to demonstrate that tool performance is superior to alternative processes and/or practices. State which performance metrics (3.2.1) were used and what you have determined to be acceptable levels of accuracy for your intended use (3.2.2). (3.2.3) Methodologists: state whether evaluation / validation studies are publicly available, and where (e.g. have they been published in a peer reviewed journal or were they preprinted or similar (i.e., shared but without quality assurance processes)?). Consider whether: • the methods are reported in sufficient enough detail to be transparent and replicable • there has been any contamination between the training, testing and validation datasets • it supports the task it claims to perform (3.2.4) Methodologists: Is it possible to replicate the validation study (or studies) independently? (3.2.5) Methodologists should document and assess issues concerning the stability of the system. This includes, for example:

· Whether the system has data security and privacy procedures appropriate for intended use ? (including retention and deletion). · Is the system run on external servers, or can it be run locally? · Is it currently available? · Does it support the languages you need (including English, where relevant)? · Does the system have a track record or plan to suggest it will be available and maintained?
(3.3) Both: determine whether tool performance is at least as accurate as current processes, or accurate enough against expected benchmarks (depends on the AI system).
(3.4) Both: can the performance of the tool change over time? For example, does it depend on 3rd party models, APIs or data sources that can be updated independently? State how end-users can monitor performance (3.4.1) and report issues, and what triggers should prompt re-validation or suspension of use. For example, if the system does not perform as expected, or whether biases are identified, there should be communication channels for reporting this to the AI tool developer. Specify how humans are (or are not) in control over the tool's inputs and outputs, how they can override outputs, and how they are able to monitor and verify its operation (3.4.2). (3.4.3) The AI development team should specify the procedure for releasing new versions, including version numbering, a publicly available changelog, advance notice to users, and state how any changes in performance will be communicated, validated and (if needed) rolled back.
(4) Usability and user capability
(4.1) Methodologists: do expected users have the technical knowledge to use the tool reliably (if needed); and if not, is this remediable (4.1.1)? AI development teams should consider whether working limitations and assumptions have been sufficiently understood (4.1.2).
(4.2) AI development teams: consider whether intended users understand working limitations and assumptions, and whether they have the ability to meet the expected use conditions of the tool (4.4)? Do users have the necessary skill requirements (4.4.1), and are there any technological gaps in the tool, and have the impact of these been communicated (4.4.2)?

(4.3) Methodologists: how easily can the tool's benefits be utilised by users? In particular: · Is it relatively easy to learn / use (4.3.1)? · Are sufficient training materials and other documentation available (4.3.2)? · Is adequate user support available (4.3.3)? · Are we aware of user complaints and how they were handled (4.3.4)? · Can the tool be used with current methods and systems (4.3.5)? · What are the costs involved in implementing / integrating the tool (4.3.6)?
(4.5) Bearing in mind the results of the above, including costs, would we (evidence synthesists or organisation) use this tool for our work? i.e. Is this a potentially valuable tool?
(Risk assessment B): methodologists and development teams should consider carefully likely negative outcomes for users and affected groups (B.1) if the tool is used. Methodologists should conduct an impact assessment of tool adoption (B.1.1), including plausible failure modes, who may be harmed, and the severity and likelihood of harms. Development teams should communicate clearly whether (and/or how) user data will be used to train, fine-tune, or improve the tool (B.1.2). Methodologists should develop a plan to mitigate any risks (B.2), including monitoring, audit checks, and an incident response and escalation pathway. Both roles should ensure that liabilities for harm, performance, and risk communication are documented (B.3).
(5) Transparency, licenses, availability & documentation
(5.1) AI development team: Include a full declaration of interest about whether the tool is produced by a commercial organisation that can gain financially from use of the tool; financial gain could include selling it, licensing it, making it available through paid subscription, etc. Include also a statement which specifies your sources of funding. i.e. how the initial development of the tool was paid for.
(5.2) AI development team: provide full and clear terms of use. Explicitly mention whether the AI system: · Will claim ownership of uploaded content or impose limitations on its use.

· Will gain rights to use the uploaded content for distribution, repurposing, or adaptation to train the AI system. · Will impose any limitations on the use of the AI system's outputs · Has privacy features that allow users to disable data collection for training and turn off or delete history or interaction tracking. · Has any restrictions on its use or licensing considerations, such as which countries it can be used in. · Has associated costs for use.
(5.3) Methodologists should state whether the code for the tool is accessible, and what software license applies to using it. (5.4) They should consider IP and whether there are any other ethical or other legal considerations, e.g. privacy, etc. for users or organisations. For closed source / hosted tools, state any conditions set by its license (5.4.1) and whether the terms of use are acceptable (5.4.2).
(5.5) Are you confident this is a responsible handover? Summarise why, and state any conditions for use (e.g. required mitigations, contexts where the tool should not be used, and when reassessment is needed).

Supplementary file 2: example of completed responsible handover tool

What type of handover is this?
Who is completing this form? For example, is it a tool development team handing a tool over to the evidence synthesis community for their use? Is it an evidence synthesis methodologist reporting the results of an evaluation and recommending a tool for use in a specific way? Or is it the result of collaboration between a tool development team and an evidence synthesis methodologist, who are jointly reporting their understanding about the suitability of a tool for use? Record the tool name and version (including model version, where relevant) and the date this form was completed. If any question cannot be answered, record this explicitly (for example, 'Unknown' or 'Not disclosed') and treat missing information as a potential risk signal. Schedule a reassessment date (e.g. annually or when major version updates occur) and record this in the form.
This form is the joint work of information retrieval methodologists at an evidence synthesis organisation (Cochrane), and an AI development team.
(1) What is the purpose of the tool?
Methodologists and AI development teams: agree a statement that defines how the tool should be used (1.1) in an evidence synthesis project. . Record the tool name, version, and deployment context (platform, local installation, API, etc.). If the AI tool uses generative AI, state the model provider and model version, and (where relevant) the prompt set or template version. Then, jointly consider whether there are contextual limitations (1.1.2), informed by input from the AI development team, and whether the intended context of use matches that originally envisaged (1.1.1). State whether the use case is to replace an existing task or method, or if it is to be used in parallel with current methods. If the latter, state which existing task or method this new approach should complement or be used alongside, and what human is required.
1.1: The system estimates the likelihood of a given study report describing a Randomized Controlled Trial (RCT), using only the title and abstract. This allows studies that are very unlikely to be RCTs to be automatically screened out during the development of systematic reviews for which this is desirable. The RCT classifier is designed for use as a binary classifier. The cut-point is set to achieve ≥99% recall (to identify at least 99% of reports of RCTs). Within Cochrane, the classifier is implemented as a step in the Screen4Me workflow, with all studies not excluded at this step being screened by a human before an inclusion decision. See figure from Thomas et al 2021. Deployed in this way, the RCT classifier aims to significantly reduce manual screening of titles and abstracts, but does not completely replace the need for some manual screening. 1.1.1 / 1.1.2: Accessible through the Cochrane Register of Studies as part of the Screen4Me workflow. Also embedded within the EPPI-Reviewer application (versions 4+) and within the screening features of the Covidence web application. All of these contexts of use match the original vision by the developers, with no additional contextual limitations introduced. General limitations are communicated to end-users in all three contexts.
(1.2) Methodologists: what interactions have you had with the tool development team? Describe the level of collaboration and any support in its development, if any.

1.2: The tool development team and Cochrane, including methodologists from the Information Retrieval Methods group, worked closely to define meaningful evaluation metrics and ensure the tool was accessible, via the Cochrane Register of Studies, to the Cochrane community.
(1.3) AI development teams: describe the development history of the tool briefly. This should include methods or approaches used to safeguard accuracy in the output, a history of data sources, errors discovered, and modifications made. Indicate whether there are details about how the model has developed over time and what future updates are planned.
1.3: The main paper (Thomas 2021) does not report any previous iterations of the classifier or historical data sources that were not ultimately used. It's unlikely that there are any unused data sources given the scarcity of high-quality labelled datasets for this classification task. Expertise gained by the authors through previous machine learning classification tasks is likely to have contributed to the choice of model and training and calibration processes. The approach taken by the development team was to prioritise recall over precision, calibrating the classifier to achieve 99% recall. The tool was developed in three stages: Training, calibration and validation. Each stage used a high quality, independent data set. To ensure suitable and safe deployment the development team and the Cochrane team consulted with the Cochrane Information Retrieval Methods Group. Further detail can be found in Thomas 2021. The RCT classifier has been updated since it was first deployed. The update included an assessment of whether re-calibration of the class probabilities was warranted, with the aim of improving precision without sacrificing recall. The new version was deployed once the assessment was complete and recall was validated at $\geq 99\%$.
(Risk assessment A): taking into account the answers given to the previous questions, consider the benefits and risks of using the tool. Evidence synthesis methodologists should consider how the tool will assist in the review process (A.1), whether it is clearly better than alternatives (A.1.1) (for example, whether it outperforms other AI tools or manual processes), and what trade-offs are introduced (time, cost, bias, transparency, sustainability). Also detail how the tool meets applicable local, national or cross-border, legal and ethical requirements (A.1.3). For example, confirm that the tool does not retain copyrighted material for model improvement, as evidence synthesis teams are rarely granted rights to permit this when downloading papers from publishers. Assign an overall risk level (low/medium/high) and record key mitigations required for use. Specify the level of autonomy (A.2) that the tool should have, including which decisions it is permitted to make without human sign-off, what human oversight is required and how users can override or audit outputs (including an audit trail). AI development teams should provide information on the tool's environmental impact (A.3) and their efforts to minimise its carbon footprint (for example, energy use, model size, data centre information, and any efficiency measures). Where estimates are not available, record this and outline plans to measure and report it.
A.1: The RCT classifier should always be used intentionally by the authors of a systematic review with a full understanding of its limitations, with the authors continuing to assume ultimate responsibility for their published work (RAISE 1).

A1.1: The classifier has been demonstrated to reduce title-abstract screening burden for RCT-based health-related evidence syntheses, having saved the Cochrane Crowd from needing to check 185,039 records manually (41% of all records) during the 2018 calendar year. The performance of the classifier has been validated using a subset of all possible RCTs related to health/healthcare. Users of the classifier should consider whether this validation dataset is representative of the studies they are aiming to classify, before assuming that decisions made by the classifier are valid for their particular use case.

A.2: Records with a sufficiently high likelihood of being RCTs are always manually reviewed by the user, so screening should never be fully autonomous. Records that fall below the pre-specified cut-point are automatically discarded. The chosen cut-point ensures $\geq 99\%$ recall and was set conservatively e.g. by using a more challenging dataset of older study reports for calibration. Users nonetheless have access to the records that are discarded and can check them at any point.

A1.3: Training and calibration were performed on title and abstract data with full permission of the database provider, ensuring compliance with publisher terms.

All source code and datasets are available on request to the extent allowable by the terms of publishers and bibliographic databases, supporting reproducibility and auditability.

The tool processes bibliographic records only, which contain no personal or sensitive data.

A.3: In terms of environmental impact, the tool is not based on the use of large language models. It is a support vector classifier and its environmental impact in terms of energy use and carbon emissions would be small (classification of 1000 records would roughly equate to writing and sending a single email)

(2) Where have the training, testing and validation data come from?

(2.1 & 2.2) Methodologists: state **what data were used to train / test / validate the tool**. Specify how they can be accessed and consider whether training dataset was adequately sized and sufficiently broad to cover the stated topics and domains adequately. **Are the data available to you?** (2.1.1 / 2.2.1) Also specify any licensing, permissions or access constraints that affect whether training and validation data can be shared or reused.

Consider:

- Are the sources of training and validation data clear and transparent (including rights and permissions for reuse)?
- Is the currency of the data adequate for your needs?
- Is the age of the data adequate for your needs?
- How data were labelled or annotated as part of building the model (if any)

Determine:

- Power: Is the corpus large enough for your work (is it adequately sized and not skewed towards certain topics)?
- Comprehensiveness: Does it contain the breadth of topics applicable? Are there gaps in the scope and domain, sources, geography, language, methods, publication types, etc.?

Coverage / currency: Is the corpus frequently updated? Are there gaps for older and newer information?

Source(s) of training data

The classifier was trained on study reports retrieved from Embase for inclusion in CENTRAL between January 2014 and July 2016.

The training data was labelled using the citizen science platform Cochrane Crowd. Each report was labeled by multiple members of Crowd, with the final label being determined by an agreement

algorithm. The recall and precision for identifying RCTs had both previously been found to exceed 99% ([Noel-Storr 2021](#)). The records assessed by the Crowd had been identified through a sensitive search for RCTs.

The Clinical Hedges dataset was then used for calibration of the classifier. This consists of 49,028 title–abstract records, with the associated full-text manually checked to ascertain with confidence whether each reported an RCT. The study reports in this dataset were older, published during 2000 and 2001, and so are therefore less likely to follow a consistent reporting structure in accordance with CONSORT. By using a more demanding dataset for calibration, the risk of an overconfident classifier was minimized.

Study reports with titles of fewer than 15 characters or abstracts of less than 400 characters were considered to contain insufficient information for machine classification and so were removed from both datasets.

Source(s) of testing data (if relevant)

To validate recall of the classifier on its primary intended use case (screening of studies for Cochrane reviews), a validation dataset was created made up of 94,305 reports of 58,283 studies assumed to be almost all RCTs given their inclusion in 4,296 Cochrane reviews with strict study eligibility criteria.

Given that this dataset contained only the positive ‘is RCT’ class, it was used only for validating the recall of the classifier and not the precision. The precision of the classifier at the selected calibration threshold was estimated using the calibration dataset mentioned above.

As for the training and calibration datasets, study reports with very short titles or abstracts were excluded.

Currency and age of the training and testing data

Currency of the data used to train and test the classifier is an important consideration here. Reporting standards for RCTs have improved since the advent of CONSORT so a classifier built on post CONSORT titles-abstract records might perform differently when presented with pre-CONSORT records. In the case of the RCT classifier the currency of the datasets used to train, calibrate and test the classifier was adequate. The training data was mainly made up of references with publication dates between 2014-2016. The calibration data was older, with most records published around 2000, and in the final test data, publication dates ranged widely spanning a 60-year period. The performance of the classifier on test dataset helped to confirm that the RCT classifier could work well on records from both pre- and post-CONSORT reporting guidance.

2.1.1 / 2.2.1: Availability of data

The training, calibration and validation datasets are available upon request from the developers of the classifier under the terms imposed by publishers and database providers.

The methodologists completing this form were heavily involved in finding the data used to develop and the evaluate the classifier and so have ready access to the datasets.

(2.3 & 2.4) AI tool developer: state whether the data used for training, testing and validation (where appropriate) **match the use case**. Do the training / development data match the intended use? Consider the strengths and limitations of the data in relation to the domains and breadth of the data used (2.3.1), including what scenarios it is best suited to and when it may be inappropriate to use, including scope and domains, sources (e.g. open access only, PubMed only), geography, language, methods, etc. Confirm that there is no contamination between datasets that were used for training, testing and validation.

2.3, 2.4: The training and testing data match the intended use case of supporting identification of RCTs for Cochrane reviews. The training data was derived from one of the main bibliographic sources using a highly sensitive validated search for RCTs. The testing data was made up of references to included studies from all available RCT based Cochrane reviews.

2.3.1: However, three potential limitations should be noted:

1. The datasets were all composed of English language title-abstract records
2. Title-only records, or records with fewer than 400 characters were excluded from the three datasets
3. The datasets were all made of title-abstract records from either journal articles or conference publications. Other record types were (for example, trial registry records) were not included

The performance of the classifier cannot be guaranteed for records that meet any the above three criteria.

There was no contamination between the training dataset and those used for calibration and validation.

(2.5) Both: state whether **the data are likely to be biased**. Consider what biases or equity issues there may have been in the system's training data and strategies for users to help mitigate potential issues due to biases, e.g. were data obtained from published sources only, or open sources only or skewed towards certain topics. Additionally, state whether the corpus was sufficiently large. Describe what steps were taken to ensure training data are representative of intended use case.

Strength - Quality of data: Three large, independent datasets (in the order of magnitude $10^4 - 10^5$ records) were used to train, calibrate and validate the classifier respectively. Availability of such high quality data at volume is an advantage when developing machine learning classifiers to reduce the risk of overfitting on the training data. In addition, many 'obvious' non-RCT reports and 'obvious' RCT reports were omitted from the training data, meaning the training data represented the 'grey area' well, with labels available for a large number of difficult records.

Limitation - Language: The classifier was trained exclusively on study reports in English, so is not expected to perform well on reports published in other languages.

Limitation - Bias against older studies: The validation described above found that older study reports, particularly those from before the widespread adoption of the CONSORT guidance, are slightly more likely to be misclassified. However, the classifier is calibrated to achieve $\geq 99\%$ recall on studies from 2000-2001, a time period in which CONSORT was not well-adopted, and so the number of studies lost is still expected to be very low, even when working with older studies.

(3) Is the AI tool validated and performing sufficiently well for use?

(3.1) Methodologists and development teams: **how do you measure the AI tool's performance, and is it being monitored?**

State the method(s) / protocol(s) used (include URL or attach) and confirm:

- *Performance metrics measured and results for each*
- *What exclusion criteria were applied*
- *Information about the types and categories of data and detail any potential biases*
- *Any processes used to clean the data, including handling missing data*

- For generative AI and large language models, state how prompts were developed and evaluated, ensuring the full prompt text is recorded in detail. Also record the model provider and version, key generation settings (e.g. temperature), and any retrieval or tool-use configuration.

State who conducted the validation(s), when, and whether they were independent of the AI tool developers. (3.1.1)

3.1: The classifier's performance has been measured during its initial development, subsequent updates, and in multiple independent validations conducted by third-parties, such as the example given below:

INDEPENDENT VALIDATION

Date(s) of validation

August 2025

Who conducted the validation(s)

Independent of the developers and their validation described above, [Moberg and Gornitzki](#) of the Swedish Agency for Health Technology Assessment and Assessment of Social Services validated the RCT classifier.

Method(s) / protocol(s) (include URL or attach) and confirm:

<https://www.cambridge.org/core/journals/research-synthesis-methods/article/combining-search-filters-for-randomized-controlled-trials-with-the-cochrane-rct-classifier-in-covidence-a-methodological-validation-study/04406772BE26A14F77510D1D539E2765>

Validation of recall: A new dataset was populated with RCTs (n=1,103) published between 1970 and 2024, from evidence syntheses published by the authors' institution. References for these studies were imported into Covidence so that the classifier could be used therein. Recall was then calculated as the number of studies classified as a possible RCT as a proportion of the total number of RCTs in the dataset. Recall was calculated for the RCT classifier used in isolation, and when used in combination with an upstream search filter for RCTs.

Assessment of utility (via specificity): Searches for three evidence syntheses by the authors' institution were replicated on both MEDLINE and Embase. The authors recorded the number of study reports requiring manual screening (NNR) when using a) no automation b) the RCT classifier c) the RCT classifier plus an upstream RCT search filter.

Source(s) of data

The dataset used in the validation of recall was generated using the relative recall method, from 47 reports published by the Swedish Agency for Health Technology Assessment and Assessment of Social Services. The 1,103 RCT study reports in the dataset are listed in Appendix 2 of the linked paper, and were published between 1982 and 2024. Without full access to the training data used to train the RCT classifier, independence between the training set and this validation set cannot be confirmed (note that this validation set is separate to the validation set used during development, which was independent of the training set)

Results - Validation of recall

	Recall	No of missed RCTs
Cochrane RCT Classifier in Covidence	99.64%	4
Cochrane sensitivity-maximizing RCT filter in Ovid MEDLINE	98.73%	14
Cochrane Embase RCT filter for Embase.com	98.46%	17
MEDLINE RCT filter AND RCT classifier combined	98.46%	17
Embase RCT filter AND RCT classifier combined	98.28%	19

Table 1: Reproduced from Moberg and Gornitzki

The recall of the classifier was 99.64%, marginally higher than the recall of RCT search filters used on either MEDLINE or Embase.

Results - Assessment of utility (via specificity)

	Original search strategy without RCT						With RCT FILTER						With RCT CLASSIFIER						With RCT FILTER and RCT CLASSIFIER					
	FILTER/RCT CLASSIFIER						With RCT FILTER						With RCT CLASSIFIER						With RCT FILTER and RCT CLASSIFIER					
				Recall of included references found by						Recall of included references found by						Recall of included references found by								
	No. of search hits	No. of included references retrieved	NNR	No. of search hits	original search strategy	NNR	No. of search hits	original search strategy	NNR	No. of search hits	original search strategy	NNR	No. of search hits	original search strategy	NNR									
SBU Assessment 337: Internet-delivered psychological treatment versus other available treatment options for common mental disorders																								
Ovid MEDLINE	10,083	20	504	5,264	100%	263	6,340	100%	317	4,373	100%	219												
Embase.com	11,603	17	683	4,597	100%	270	7,356	100%	433	4,013	100%	236												
SBU Assessment 372: Treatment and social support for adults with co-occurring addictive and psychiatric disorders																								
Ovid MEDLINE	91,324	88	1,038	25,016	100%	284	33,817	100%	384	11,609	100%	132												
Embase.com	71,396	77	927	13,274	100%	172	19,703	100%	256	6,040	100%	78												
SBU Policy support 379: Treatment and rehabilitation of post-COVID-19 and other post-infectious conditions																								
Ovid MEDLINE	16,149	39	414	2,993	100%	77	4,836	100%	124	1,337	100%	34												
Embase.com	18,131	36	504	2,914	100%	81	6,021	100%	167	1,230	100%	34												

Table 2: Reproduced from Moberg and Gornitzki. NNR = number of study reports requiring manual screening

The RCT classifier screened out an average of 56.7% of non-RCT studies when searching MEDLINE, and 58.6% of non-RCT studies when searching EMBASE. There was no loss of sensitivity, with 100% recall for both test cases. The classifier was consistently outperformed with respect to specificity by the RCT search filters, with the optimal specificity being achieved by using the classifier and a search filter in combination.

(3.2) Methodologists, state the **comparator** (if relevant), and how this is able to demonstrate that tool performance is superior to alternative processes and/or practices. State which **performance metrics** (3.2.1) were used and what you have determined to be **acceptable levels of accuracy** for your intended use (3.2.2).

(3.2.3) Methodologists: state whether evaluation / **validation studies are publicly available**, and where (e.g. have they been published in a peer reviewed journal or were they preprinted or similar (i.e., shared but without quality assurance processes)?). Consider whether:

- the methods are reported in sufficient enough detail to be transparent and replicable
- there has been any contamination between the training, testing and validation datasets
- it supports the task it claims to perform

(3.2.4) Methodologists: Is it possible to **replicate the validation study** (or studies) independently?

(3.2.5) Methodologists should document and assess issues concerning the **stability of the system**. This includes, for example:

- Whether the system has data security and privacy procedures appropriate for intended use ? (including retention and deletion).
- Is the system run on external servers, or can it be run locally?
- Is it currently available?
- Does it support the languages you need (including English, where relevant)?

Does the system have a track record or plan to suggest it will be available and maintained?

3.2: The comparator in the validation study was manual screening, with and without use of an upstream search filter.

3.2.1: The independent validation sought to assess both recall and workload reduction (proportion screened out) as the two main performance metrics.

3.2.2: The classifier achieved over the required minimum recall of 98.5%. There is not an accepted level of workload reduction but achieving over 50% (which the classifier did in the independent validation study) is considered good.

3.2.3: The [main validation study](#) is in an open access, peer reviewed journal.⁶¹ The paper describes the development, calibration and validation of the classifier in detail, and makes it clear that there was not contamination between the three datasets used. The results support the task it claims to perform. The independent validation study detailed above is also open access in a peer-reviewed journal.⁶²

3.2.4: The study has been reported in sufficient detail to be replicated.

Appeals and feedback pathway

Feedback and concerns can be provided via emailing support@cochrane.org - All feedback and concerns will be directly passed onto the tool developers.

3.2.5: The RCT Classifier is accessible from the Cochrane Register of Studies, EPPI Reviewer and Covidence. The system does not store any sensitive or personal data. All are cloud-based systems and, while the RCT classifier could be deployed locally, this would need some technical work. All systems have been maintained with minimum downtime for maintenance for the past 7-8 years.

(3.3) Both: determine whether tool performance is at least as accurate as current processes, or accurate enough against expected benchmarks (depends on the AI system).

Validation of the classifier required it to achieve a recall of $\geq 99\%$ which is higher than the 98.5% recall achieved by Cochrane's Highly Sensitive Search Strategy which is widely adopted as a standard process.

(3.4) Both: can the performance of the tool **change over time**? For example, does it depend on 3rd party models, APIs or data sources that can be updated independently? State how **end-users can monitor performance** (3.4.1) and report issues, and what triggers should prompt re-validation or suspension of use. For example, if the system does not perform as expected, or whether biases are identified, there should be communication channels for reporting this to the AI tool developer. Specify how humans are (or are not) in control over the tool's inputs and outputs, how they can override outputs, and how they are able to monitor and verify its operation (3.4.2).

(3.4.3) The AI development team should specify the **procedure for releasing new versions, including version numbering, a publicly available changelog, advance notice to users**, and state how any changes in performance will be communicated, validated and (if needed) rolled back.

3.4: Each version of the classifier is stable, with its performance not dependant on any third-party services.

3.4.1: End-users can conduct their own validations of the classifier's performance using their own datasets of RCT report. If any performance issues or biases are identified, these should be reported to the tool developers via support@cochrane.org.

3.4.2: Humans control which title-abstract records are used as inputs to the tool and always remain responsible for deciding how to use its output. They can verify its operation by checking the number of records discarded does not exceed the expected proportion, and by spot-checking

a small volume of decisions manually. They may also wish to carry out a full validation of performance on their own dataset, as described above, if they feel the validations published to-date are insufficient.

3.4.3: New versions will be released periodically. Prior to a new version release, checks will be made to ensure that the new version performs as expected (remaining at $\geq 99\%$ recall and with improved precision when compared to older version/s).

(4) Usability and user capability

(4.1) Methodologists: do expected users have the **technical knowledge to use the tool reliably** (if needed); and if not, **is this remediable** (4.1.1)? AI development teams should consider whether **working limitations and assumptions have been sufficiently understood** (4.1.2).

Very little technical knowledge is required to use the RCT classifier effectively. The only requirement is an ability to ensure that the records sent to the classifier are suitable in that they lie within the domain for which the classifier was designed. This is clearly communicated to end-users of EPPI-reviewer and Covidence via user documentation. The binary nature of the classifier means that users are not required to interpret scores or derive their own cut-point.

(4.2) AI development teams: consider whether intended users **understand working limitations and assumptions**, and whether they have the ability to meet the **expected use conditions** of the tool (4.4)? Do users **have the necessary skill requirements** (4.4.1), and **are there any technological gaps in the tool, and have the impact of these been communicated** (4.4.2)?

The RCT classifier is straightforward to use as calibrated class probabilities and cut-off points are presented automatically in the three contexts of use (EPPI Reviewer, Covidence and Cochrane Register of Studies). Each tool has guidance and support available for users.

(4.3) Methodologists: **how easily can the tool's benefits be utilised by users?** In particular:

- Is it relatively easy to learn / use (4.3.1)?
- Are sufficient training materials and other documentation available (4.3.2)?
- Is adequate user support available (4.3.3)?
- Are we aware of user complaints and how they were handled (4.3.4)?
- Can the tool be used with current methods and systems (4.3.5)?

What are the costs involved in implementing / integrating the tool (4.3.6)?

The RCT classifier is straightforward to use as the cut-point used is set by default and deployed in three systematic review tools (EPPI Reviewer, Covidence and CRS). Each tool has guidance and support available for users.

:

- [Cochrane Register of studies](#)
- [Covidence](#)
- [EPPI-Reviewer](#)

(4.5) Bearing in mind the results of the above, including costs, would we (evidence synthesists or organisation) use this tool for our work? i.e. **Is this a potentially valuable tool?**

This tool is valuable in helping to reduce the screening burden. However, it does not eliminate that burden entirely and its performance, in terms of workload reduction, will vary depending on factors such as prevalence of RCTs in the domain area and sensitivity of the search used to identify the records. Screen4Me is routinely used in many Cochrane reviews seeking to include randomised controlled trials.

(Risk assessment B): methodologists and development teams should consider carefully **likely negative outcomes for users and affected groups** (B.1) if the tool is used. Methodologists should conduct an **impact assessment of tool adoption** (B.1.1), including plausible failure modes, who may be harmed, and the severity and likelihood of harms. Development teams should communicate clearly whether (and/or how) **user data will be used** to train, fine-tune, or improve the tool (B.1.2). Methodologists should develop a **plan to mitigate any risks (B.2)**, including monitoring, audit checks, and an incident response and escalation pathway. Both roles should ensure that **liabilities for harm, performance, and risk communication** are documented (B.3).

B.1: Outcome

Relevant studies are missed due to inappropriate use of classifier outside of domain intended by developers

Users misunderstand output of classifier as containing only RCT records.

Users use classifier to screen expecting systematic reviews containing RCTs to be screened in

B.2: Mitigation

The domain in which the classifier's performance has been validated has clearly been stated in user documentation. Deployment of the classifier in any new contexts / applications should be preceded by consideration of expected use.

Any presentation of the classifier's output is accompanied by a clear explanation that contains records that *may be RCTs*

User documentation of web application clearly states that the classifier aims to screen out records that do not directly describe an RCT. Managers of the Screen4Me service also provide the same clarification to service users.

B1.1: An impact assessment is outlined in [Thomas 2021](#). The recall of the classifier upon a validation set of 44,007 sufficiently reported studies from 4296 Cochrane reviews was 99.5%. There is a possibility that the 0.5% of studies that were lost may have influenced the results of up to 4% of the reviews; no studies from the remaining 96% of reviews were lost. However, the studies that were lost were overwhelmingly older reports, and so the likelihood of a review's conclusions being affected is considered very low.

B1.2: User data will not be used to train the classifier.

B.3: Liabilities

Liability for Harm

The classifier is intended for use within validated workflows (e.g. Cochrane Screen4Me) and should not be applied outside its intended scope without additional validation. Users retain ultimate responsibility for review accuracy. Developers disclaim liability for decisions made solely on classifier outputs without human oversight. Known limitations include: Reduced performance on non-English records and older studies. Potential bias due to training data sourced primarily from Embase and Cochrane datasets.

Liability for Performance

Performance metrics (recall, precision, calibration) have been validated internally and independently; results are publicly available. The classifier is calibrated to maintain $\geq 99\%$ recall under specified conditions. Precision may vary depending on dataset characteristics. Developers commit to monitoring performance and communicating any changes following version updates.

Liability for Risk Communication

Developers provide full documentation of:

Intended use and limitations. Training, calibration, and validation datasets. Version history and update procedures.

Feedback and issue reporting channels are provided via support@cochrane.org. Users are advised to spot-check discarded records periodically to verify performance is as-documented, and review all relevant documentation before using the classifier in any deployment context.

No rights are claimed over user-uploaded content; outputs may be freely used with appropriate attribution.

(5) Transparency, licenses, availability & documentation

(5.1) AI development team: Include a **full declaration of interest** about whether the tool is produced by a commercial organisation that can gain financially from use of the tool; financial gain could include selling it, licensing it, making it available through paid subscription, etc. Include also a statement which specifies your sources of funding. i.e. how the initial development of the tool was paid for.

The tool is not produced by any commercial organisation that could gain financially from it.

Covidence may gain financially from inclusion of the classifier as an attractive feature of their software offering, however this was offered free-of-charge to Covidence in order to benefit Cochrane authors.

Development funding sources

The developers were funded by Cochrane via Project Transform; Australian National Health & Medical Research Council (Partnership Project grant APP1114605); U.S. National Library of Medicine (Award 2R01-LM012086-05).

Some of the developers were also supported by a Medical Research Council (UK) fellowship (MR/N015185/1), and by the National Institute for Health Research (NIHR) Collaboration for Leadership in Applied Health Research and Care North Thames at Barts Health NHS Trust.

(5.2) AI development team: provide full and **clear terms of use**. Explicitly mention whether the AI system:

- Will claim ownership of uploaded content or impose limitations on its use.
- Will gain rights to use the uploaded content for distribution, repurposing, or adaptation to train the AI system.
- Will impose any limitations on the use of the AI system's outputs
- Has privacy features that allow users to disable data collection for training and turn off or delete history or interaction tracking.
- Has any restrictions on its use or licensing considerations, such as which countries it can be used in.
- Has associated costs for use.

The classifier is open source and can be freely used by anyone with access to the training data. Use of the classifier is possible within [EPPI-Reviewer](#), [Covidence](#) or the [Cochrane Register of Studies](#). In addition, access via Cochrane's [Screen4Me](#) service is available to authors of Cochrane reviews free of charge.

(5.3) Methodologists should state whether the **code for the tool is accessible**, and what **software license applies** to using it.

(5.4) They should **consider IP** and whether there are any other ethical or other legal considerations, e.g. privacy, etc. for users or organisations. For closed source / hosted tools, state any **conditions set by its license** (5.4.1) and whether the **terms of use are acceptable** (5.4.2).

Source code: A high-level outline of how the classifier was developed is given in [Thomas 2021](#). The overall classifier is an ensemble of two support vector classifiers. The source code for both of these is [available](#) publicly, though assembling them would need some technical expertise.

Training dataset: Available upon request from support@cochrane.org

Terms of use

The RCT classifier does not store any titles and abstracts fed into it, and no rights or ownership of these are claimed at any point. Users of the classifier are not limited in how they use the classifier's outputs, but should credit the developers of the classifier.

(5.5) Are you **confident this is a responsible handover**? Summarise why, and state any conditions for use (e.g. required mitigations, contexts where the tool should not be used, and when reassessment is needed).

Yes, and the decision would be to proceed when use of this tool as the validation evidence is strong, transparency is high, and risks are well-understood and reasonably manageable.